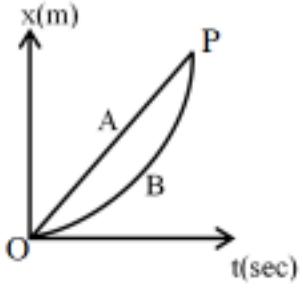
PHYSICS

1) A projectile is launched at an angle 'α' with the horizontal with a velocity 20 ms^{-1}. After 10 s, its inclination with horizontal is 'β'. The value of tanβ will be (g = 10 ms^{-2}) :-

(1) $\tan\alpha + 5\sec\alpha$
(2) $\tan\alpha - 5\sec\alpha$
(3) $2\tan\alpha - 5\sec\alpha$
(4) $2\tan\alpha + 5\sec\alpha$

2) Consider the given position (x) - time (t) curves for two cars A and B.



Correct statement is :-

(1) During the motion relative velocity of cars is zero only for once.
(2) Average velocity of both cars is equal from O to P.
(3) Velocity of car A is constant.
(4) All of above

3) A particle crossing the origin of the coordinates at time t = 0, moves in the xy plane with a constant acceleration β in the y-direction. If its equation is $y = \alpha x^2$ (α is a constant), then its velocity component in the x-direction is :-

(1) $\sqrt{\frac{\alpha}{\beta}}$
(2) $\sqrt{\frac{\beta}{\alpha}}$
(3) $\sqrt{\frac{\alpha}{2\beta}}$
(4) $\sqrt{\frac{\beta}{2\alpha}}$

4) Two particles, positioned at A(5, 3) and B(7, 3), are moving with constant velocity $2\hat{i} + 3\hat{j}$ and $x\hat{i} + y\hat{j}$ respectively. After 2 s they collide, then the values of x and y are respectively

(1) 2, 2
(2) 1, 3

(3) 3, 2

(4) 1, 1

5) A small block of mass m is kept on a rough inclined surface of inclination θ fixed in a elevator. The elevator goes down with a uniform velocity v and the block does not slide on the wedge. The work done by the force of friction on the block in time t will be :-

(1) zero

(2) –mgvt $\cos^2\theta$

(3) –mgvt $\sin^2\theta$

(4) mgvt $\sin^2\theta$

6) A particle moves with a velocity $\vec{v} = (5\hat{i} - 3\hat{j} + 6\hat{k})$ m/s under the influence of a constant force $\vec{F} = (10\hat{i} + 10\hat{j} + 20\hat{k})N$. The instantaneous power applied to the particle is :-

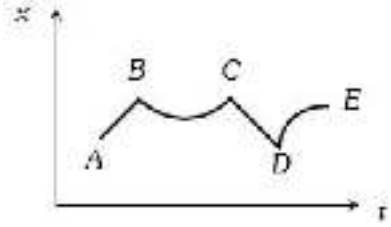
(1) 200 J/s

(2) 40 J/s

(3) 140 J/s

(4) 170 J/s

7) Figure shows the displacement of a particle going along the X-axis as a function of time. The force acting on the particle is zero in the region :



(1) AB

(2) BC

(3) CD

(4) Both (1) and (3)

8) A rocket is ejecting 50 g of gases per sec at a speed of 500 m/s. The accelerating force on the rocket will be..... N

(1) 125

(2) 25

(3) 5

(4) 0

9) Particles of masses m, 2m, 3m,..., nm are placed on the same line at distance L, 2L, 3L,..., nL from origin O. The distance of centre of mass from O is-

(1) $\frac{(2n+1)L}{4}$

(2) $\frac{L}{(2n+1)}$

(3) $\frac{n\left(n^2+1\right)L}{2}$

(4) $\frac{(2n+1)L}{3}$

10) A molecule in a gas container hits a horizontal wall with speed 200 ms^{-1} and angle 30° with the normal and rebounds with the same speed. Which of the following option is correct?

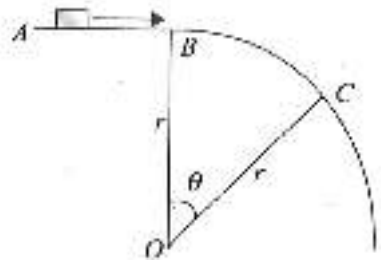
(1) Momentum is not conserved.

(2) Elastic collision occurs here.

(3) Inelastic collision occurs here.

(4) Both (1) and (2)

11) A small block slides with velocity $v_0 = 0.5\sqrt{gr}$ on the horizontal frictionless surface as shown in the fig. The block leaves the surface at point C. The angle θ in the figure is:



(1) $\cos^{-1}\frac{4}{9}$

(2) $\cos^{-1}\frac{3}{4}$

(3) $\cos^{-1}\frac{1}{4}$

(4) $\cos^{-1}\frac{4}{5}$

12) A car has to move on a level turn of radius 45 m. The coefficient of static friction between the tyre and the road is $\mu_S = 0.5$. Find the maximum speed the car can take without skidding.

(1) 36 Km/hr

(2) 18 Km/hr

(3) 54 Km/hr

(4) 108 Km/hr

13) Two gases *A* and *B* are filled at the same pressure in separate cylinders with movable pistons of radius r_A and r_B, respectively. On supplying an equal amount of heat to both the systems reversibly under constant pressure, the pistons of gas *A* and *B* are displaced by 16 *cm* and 9 *cm*, respectively. If the change in their internal energy is the same, then the ratio $\frac{r_A}{r_B}$ is equal to

(1) $\frac{4}{3}$

(2) $\frac{3}{4}$

(3) $\frac{2}{\sqrt{3}}$

(4) $\frac{\sqrt{3}}{2}$

14) We have a jar A filled with gas characterised by parameters, P V and T and another jar B filled with gas with parameters 2P, $\frac{V}{4}$ and 2T, where the symbols have their usual meanings. The ratio of the number of molecules of jar A to those of jar B is-

(1) 1 : 1

(2) 1 : 2

(3) 2 : 1

(4) 4 : 1

15) If 2 mol of an ideal monatomic gas at temperature T_0 are mixed with 4 mol of another ideal monatomic gas at temperature $2T_0$, then the temperature of the mixture is :-

(1) $\frac{5}{3}T_0$

(2) $\frac{3}{2}T_0$

(3) $\frac{4}{3}T_0$

(4) $\frac{5}{4}T_0$

16) Six identical conducting rods are joined as shown in figure. Points A and D are maintained at temperatures 200°C and 20°C, respectively. the temperature of junction B will be :-



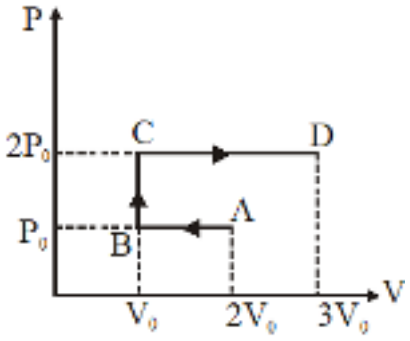
(1) 120°C

(2) 100°C

(3) 140°C

(4) 80°C

17) P-V diagram of an ideal gas is as shown in figure. Work done by the gas in the process ABCD is



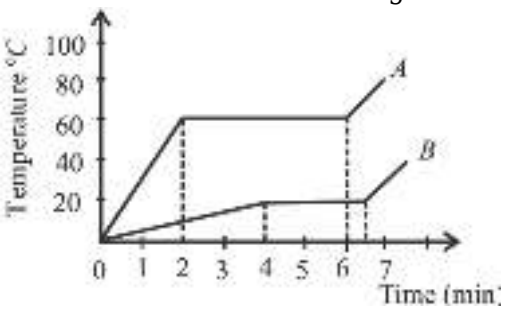
(1) $4P_0V_0$

(2) $2P_0V_0$

(3) $3P_0V_0$

(4) P_0V_0

18) Two substances A and B of equal mass m are heated at uniform rate of 6 cal s^{-1} under similar conditions. A graph between temperature and time for both is shown in figure. Ratio of heat absorbed $\left(\frac{H_A}{H_B}\right)$ by them for only fusion is :-



(1) $\frac{9}{4}$

(2) $\frac{4}{9}$

(3) $\frac{8}{5}$

(4) $\frac{5}{8}$

19) When the displacement of a particle executing SHM is one-fourth of its amplitude, what fraction of the total energy is the kinetic energy ?

(1) $\frac{16}{15}$

(2) $\frac{15}{16}$

(3) $\frac{3}{4}$

(4) $\frac{4}{3}$

20) ***Assertion :-*** In a simple harmonic motion, velocity does not change uniformly with displacement.

Reason :- The graph between velocity and displacement for a harmonic oscillator is a parabola.

(1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.

(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.

(3) Assertion is True but the Reason is False.

(4) Both Assertion & Reason are False.

21) If a spring block system has time period T and now spring is cut into 'n' equal parts, then the time period of each part connected with the same block will be :-

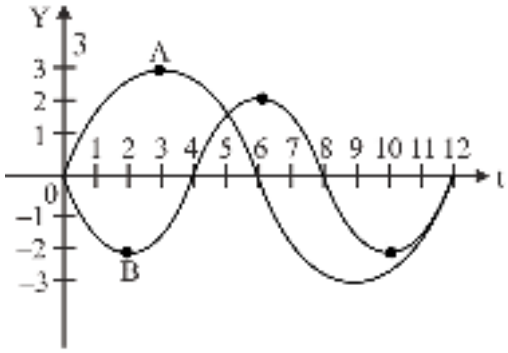
(1) $\frac{T}{\sqrt{n}}$

(2) $T\sqrt{n}$

(3) nT

(4) T

22) The displacement vs time graph for two waves A and B which travel along the same string are



shown in the figure. Their intensity ratio I_A/I_B is

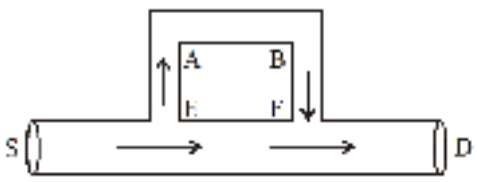
(1) $\frac{9}{4}$

(2) 1

(3) $\frac{81}{16}$

(4) $\frac{3}{2}$

23) A sound wave starting from source S, follows two paths SEFD and SEABFD. AB = □, AE = BF = 0.6 □ and wave length of wave is 6 m. If maximum sound is heard at D, then minimum value of length □ is-



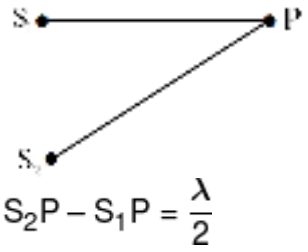
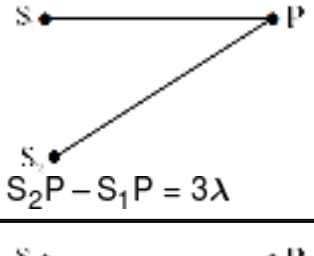
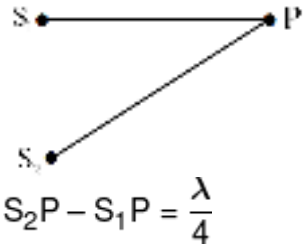
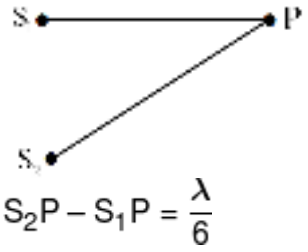
(1) 11m

(2) 6m

(3) 2.5m

(4) 5m

24) Two waves of sound having same intensity (I) and amplitude (A) interfere to produce interference at point P where resultant intensity is I_{net}, then match the following :

	Column-I		Column-II
1	 $S_2P - S_1P = \frac{\lambda}{2}$	a	$I_{net} = 2I$
2	 $S_2P - S_1P = 3\lambda$	b	$I_{net} = 4I$
3	 $S_2P - S_1P = \frac{\lambda}{4}$	c	$I_{net} = 0$
4	 $S_2P - S_1P = \frac{\lambda}{6}$	d	$I_{net} = 3I$
		e	$I_{net} = I$

(1) 1 → c, 2 → b, 3 → e, 4 → d

(2) 1 → b, 2 → c, 3 → a, 4 → d

(3) 1 → c, 2 → b, 3 → a, 4 → d

(4) 1 → c, 2 → b, 3 → d, 4 → a

25) A copper ball of mass 100 gm is at a temperature T. It is dropped in a copper calorimeter of mass 100 gm, filled with 170 gm of water at room temperature (30°C). Subsequently, temperature of the system is found to be 75°C. Then T is given by :
Given room temperature = 30°C
Specific heat of copper = 0.1 cal/gm°C

(1) 1250°C

(2) 825°C

(3) 800°C

(4) 885°C

26) A spherical ball of mass 20 kg is stationary at the top of a hill of height 100 m. It rolls down a surface to the ground, then climbs up another hill of height 30 m and finally rolls down to a horizontal base at a height of 20 m above the ground. The velocity attained by the ball is :-

(1) $40\sqrt{\frac{5}{7}}$ m/s

(2) 20 m/s

(3) 10 m/s

(4) $10\sqrt{30}$ m/s

27) A rigid spherical body is spinning around main axis without any external torque. Due to increase in temperature its volume increases by 3%. Then percentage change in its angular speed is :

(1) –2%

(2) –1%

(3) –3%

(4) 1%

28) Moment of inertia of a square frame consisting of four identical rods, about axis passing through diagonal is 24 kg m^2, then moment of inertia of one rod, about axis passing through its center and perpendicular to length will be-

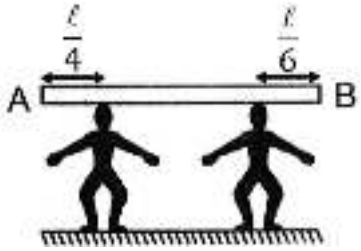
(1) 12 kg m^2

(2) 3 kg m^2

(3) 6 kg m^2

(4) 20 kg m^2

29) Two persons of equal height are carrying a long uniform wooden beam of length □. They are at distance $\frac{\ell}{4}$ and $\frac{\ell}{6}$ from nearest ends of the rod. The ratio of normal reaction at their heads $\left(\frac{N_A}{N_B}\right)$ is :-



(1) 2 : 3

(2) 3 : 2

(3) 4 : 3

(4) 3 : 4

30) The speed v of ripples on the surface of water depends on surface tension (σ), density (ρ) and wavelength (λ). The square of speed (v) is proportional to :-

(1) $\frac{\sigma}{\rho\lambda}$

(2) $\frac{\rho}{\sigma\lambda}$

(3) $\frac{\lambda}{\sigma\rho}$

(4) ρσλ

31) A student measures the period of oscillation of a simple pendulum in successive measurement, the reading turns out to be 1.93 s, 1.99 s, 2.06 s, 2.08 s and 1.95 s. A more accurate way to write the measurement with error is

(1) (2.03 ± 0.06) s

(2) (2.00 ± 0.05) s

(3) (2.00 ± 0.06) s

(4) (2.03 ± 0.1) s

32) An artificial satellite is moving in a circular orbit around the earth. The height of the satellite above the surface of earth is R. Suppose the satellite is stopped suddenly in its orbit and allowed to fall freely, on reaching earth, its speed will be :- (g is acceleration due to gravity at earth's surface, R is Radius of earth)

(1) $\sqrt{gR}$

(2) $2\sqrt{gR}$

(3) $3\sqrt{gR}$

(4) $5\sqrt{gR}$

33) With what angular velocity the earth should spin in order that a body lying at 37º latitude may become weightless.

(1) $\frac{5}{4}\sqrt{\frac{g}{R}}$

(2) $\frac{25}{16}\sqrt{\frac{g}{R}}$

(3) $\frac{5}{3}\sqrt{\frac{g}{R}}$

(4) $\frac{25}{9}\sqrt{\frac{g}{R}}$

34) Two spheres of mass 4 kg and 8 kg are placed such that their centre are 12 m away from each other. Find net gravitational force intensity at their centre of mass.

(1) $\frac{7}{16}G$

(2) $\frac{8}{7}G$

(3) $\frac{4}{7}G$

(4) $\frac{3}{8}G$

35) Two bodies, each of mass M, are kept fixed with a separation 2L. A particle of mass m is projected from the mid-point of the line joining their centres, perpendicular to the line. The gravitational constant is G. The correct statement(s) is (are) :-
(a) The minimum initial velocity of the mass m to escape the gravitational field of the two bodies is $4\sqrt{\frac{GM}{L}}$
(b) The minimum initial velocity of the mass m to escape the gravitational field of the two bodies is $2\sqrt{\frac{GM}{L}}$
(c) The minimum initial velocity of the mass m to escape the gravitational field of the two bodies is $\sqrt{\frac{2GM}{L}}$
(d) The energy of the mass m remains constant

(1) a, b
(2) b, d
(3) a, c
(4) a, d

36) An aluminium rod with Young's modulus Y = 7.0×10^{10} N/m^2 undergoes elastic strain of 0.04%. The energy per unit volume stored in the rod (in J/m^3) is:

(1) 5600
(2) 8400
(3) 2800
(4) 11200

37) **Assertion :** A gypsy car has canvas top. When the gypsy runs at high speed, the top bulges out.
Reason : According to Bernouli's theorem, total energy per unit volume of an ideal fluid in steady flow is constant i.e.
$P + \rho gh + \frac{1}{2}\rho V^2 = \text{constant}$

(1) Both Assertion and Reason are true and the Reason is correct explanation of the Assertion.
(2) Both Assertion and Reason are true, but Reason is not correct explanation of the Assertion.
(3) Assertion is true, but the Reason is false.
(4) Assertion is false, but the Reason is true.

38) The excess pressure due to surface tension inside a spherical drop is 6 unit. If eight such drops combine, then the excess pressure due to surface tension inside the larger drop is :-

(1) 3 units
(2) 6 units
(3) 12 units
(4) 48 units

39) If the terminal speed of a sphere of gold (density = 19.5 kg/m^3) is 0.2 m/s in a viscous liquid (density = 1.5 kg/m^3), find the terminal speed of a sphere of silver (density = 10.5 kg/m^3) of the same

size in the same liquid.

(1) 0.1 m/s

(2) 0.2 m/s

(3) 0.4 m/s

(4) 0.133 m/s

40) In experiment to calculate coefficient of viscosity by measuring terminal velocity, it was observed that a ball of mass $\left(\frac{\pi}{2}\right)$ gm and having radius 0.5 cm takes 5 second to fall steadily through a height of 50 cm inside a long column of liquid of density 1.2 gm/cc. The coefficient of viscosity of liquid will be given as :

(1) 4.9 poise

(2) 9.8 poise

(3) 12.6 poise

(4) 16 poise

41) A tuning fork of frequency 250 Hz is held over a resonance tube and resonance is obtained when the length of air column is 31 cm and 97 cm. The velocity of sound and radius of tube can be given by :

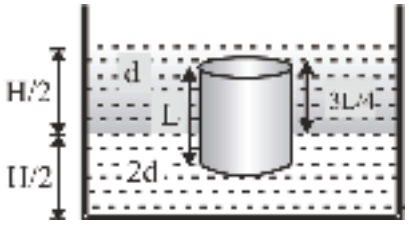
(1) 330 ms^{-1}, 3.33 cm

(2) 345 ms^{-1}, 1.1 cm

(3) 340 ms^{-1}, 2.2 cm

(4) 347 ms^{-1}, 0.6 cm

42) A homogeneous solid cylinder of length L(L < H/2) and cross-sectional area A/5 is immersed such that it floats with its axis vertical at the liquid-liquid interface with length L/4 in the denser liquid as shown in the fig. The lower density liquid is open to atmosphere having pressure P_0. Then density of solid given by :-



(1) $\frac{5}{4}d$

(2) $\frac{4}{5}d$

(3) d

(4) $\frac{d}{5}$

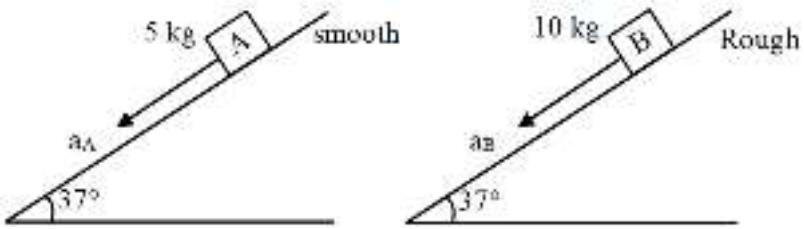
43) In a vernier callipers, 5 vernier scale divisions are equal to 4 main scale divisions. One main scale division is equal to 5 mm. A screw gauge with pitch equal to least count of above vernier callipers is designed. Number of divisions on circular scale of this screw gauge is equal to 100. Least count of this screw gauge is

(1) 0.1 mm

(2) 0.01 mm

(3) 0.001 mm

(4) 0.0001 mm

44) 

The ratio of acceleration of block A placed on smooth inclined with block B placed on rough inclined is 2 : 1. The coefficient of kinetic friction between block B and incline is :-

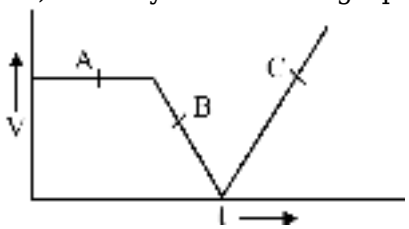
(1) 0.5

(2) 0.375

(3) 0.75

(4) 0.57

45) Velocity versus time graph of a body is shown in Fig. It explains that :-



(1) At B, force is zero

(2) At B, there is a force but towards motion

(3) At B, there is a force but it opposes the motion

(4) Forces are equal at A, B and C

CHEMISTRY

1) Identify group, Period, block of element if its divalent cation having electronic configuration $1s^2$ $2s^2$ $2p^6$ $3s^2$ $3p^6$ $3d^{10}$ $4s^2$ $4p^6$ $4d^8$:-

(1) Group–10, Period–5, Block–d

(2) Group–12, Period–4, Block–d

(3) Group–10, Period–4, Block–d

(4) doesn't exist configuration is wrong

2) If each orbital can hold maximum $4e^-$, the no. of element in 4th period will be ?

(1) 34

(2) 36

(3) 48

(4) 32

3) Find out True (T) and False (F) statement.
(i) Z_{eff} of Li^+ > Z_{eff} of B^{3+}
(ii) σ of B < σ of C
(iii) Value of Z_{eff} of mono electronic species is equal to number of protons. (iv) Z_{eff} is fundamental factor in a period to explain properties of an element.

(1) F T F T

(2) F T T T

(3) F F T T

(4) T F F F

4) Match Column-I with Column-II :

(i)	Unnilhexium	(p)	Roentgenium
(ii)	Ununnium	(q)	Dubnium
(iii)	Unnilunium	(r)	Seaborgium
(iv)	Unnilpentium	(s)	Mendelevium

(1) (i)-(s) ; (ii)-(p) ; (iii)-(r) ; (iv)-(q)

(2) (i)-(r) ; (ii)-(p) ; (iii)-(s) ; (iv)-(q)

(3) (i)-(r) ; (ii)-(s) ; (iii)-(p) ; (iv)-(q)

(4) (i)-(q) ; (ii)-(r) ; (iii)-(p) ; (iv)-(s)

5) Which of the following ionisation energy order is not correct?

(1) Mn^{+2} < Mn^{+3}

(2) N^{-3} < N

(3) Cu^+ < Cu^{+2}

(4) Ne^+ < Ne

6) Correct order of EA is ?

(1) F > Cl > Br

(2) O > Se > S

(3) S > Se > O

(4) P > S > Cl

7) **Assertion :** C_2 molecule has only π-bond.
Reason : If s-p mixing is supposed to be absent in C_2 then it will be paramagnetic.

(1) Both **Assertion** and **Reason** are true but **Reason** is NOT the correct explanation of **Assertion**.

(2) **Assertion** is true but **Reason** is false.

(3) **Assertion** is false but **Reason** is true.

(4) Both **Assertion** and **Reason** are true and **Reason** is the correct explanation of **Assertion**.

8)

In how many species the central atom has two lone pairs of electron?

XeF_4, XeF_5^-, F_2SeO_2, XeF_3^+, $XeOF_4$, $ClOF_4^+$, ICl_4^-, SCl_2, SOF_4

(1) 6

(2) 5

(3) 7

(4) 4

9) The correct order of dipole moment is:

(1) $CH_4 < NF_3 < NH_3 < H_2O$

(2) $BF_3 > CO_2 > CCl_4$

(3) $HF > H_2O > H_2S$

(4) $HF > SO_3 > SO_2$

10) Which molecule has planar structure:

(a) H_2O (b) NH_4^+

(c) XeF_2 (d) XeF_4

(e) SO_2 (f) SF_4

(g) BrF_3

(1) only a, c and d

(2) only a, c, d, e and g

(3) only d, e and g

(4) only a and c

11) Which of the following order is incorrect :-

(1) $PCl_3F_2 < PF_3Cl_2$ Dipole moment

(2) N–N < P–P bond length

(3) $PH_3 < NH_3$ % s character of l.p.

(4) $PH_3 < NH_3$ bond angle

12) Incorrect statement is/are

(a) Number of antibonding electrons in O_2^-, O_2 and O_2^{2-} are 7, 6, 8 respectively.

(b) In sp^3d^3 hybridisation ∂_{xy}, ∂_{yz} and ∂_{xz} orbital participate.

(c) MP order $NaF < MgF_2 < AlF_3$

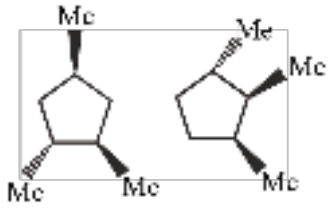
(d) In vinylacetylene C_2–C_3 overlapping is sp-sp

(1) b, d

(2) a, c

(3) a, b

(4) c, d

13) The following pair of compounds is best described as :- 

(1) Identical

(2) Diastereomers

(3) Enantiomers

(4) Position isomer

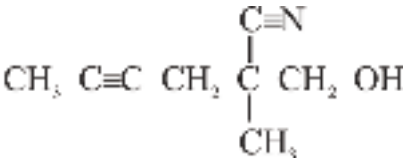
14) Identify correct statement from the following.
A. Propanal and propanone are functional group isomer
B. Ethoxy ethane and methoxypropane are metamers
C. But-2-ene shows optical isomerism
D. But-1-ene and but-2-ene are functional isomers
E. Pentane and 2, 2-dimethyl propane are chain isomers.

(1) B, C and D

(2) A, B and C

(3) A, B and E

(4) C, D and E

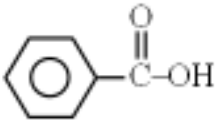
15) What is the IUPAC name of the following compound ? 

(1) 2-cyano-2-methyl hex-4-yne ol

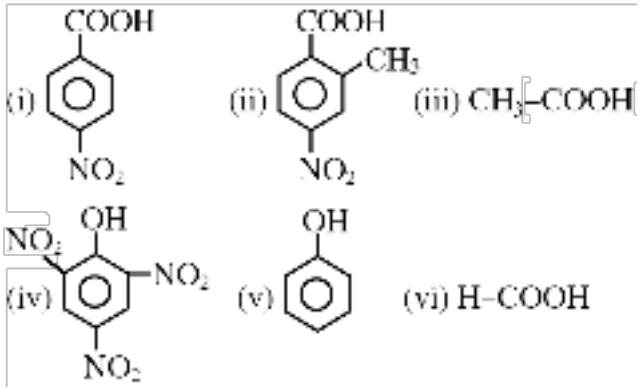
(2) 2-hydroxy methyl-2-methyl-hex-4-yne nitrile

(3) 2-hydroxy methyl-2-methyl hex-5-yne nitrile

(4) 1-cyano-2-hydroxymethyl-2-methyl-4-hexyne

16) Number of compounds which are more acidic than 

:-



(1) 2

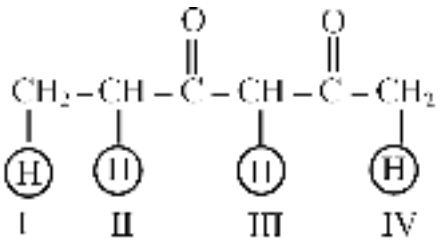
(2) 3

(3) 4

(4) 5

17)

Among the following the most acidic hydrogen is–



(1) II

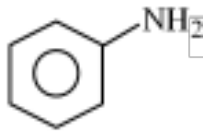
(2) I

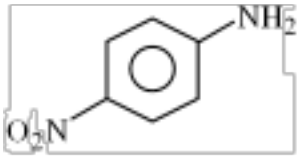
(3) IV

(4) III

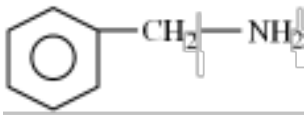
18)

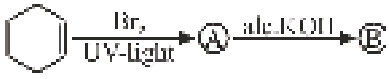
Which of the following has highest value of K_b :–

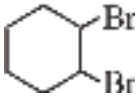
(1) 

(2) 

(3) $CH_3 – CH_2 – NH_2$

(4) 

19)  find Ⓑ :-

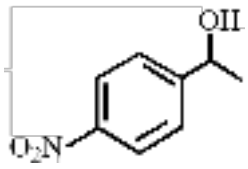
(1) 

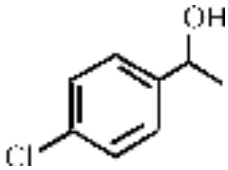
(2) 

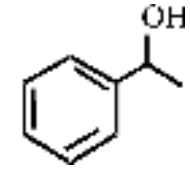
(3) 

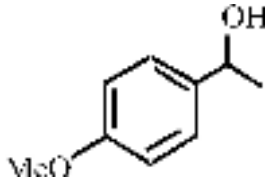
(4) 

20) Which of the following alcohols will undergo easiest dehydration ?

(1) 

(2) 

(3) 

(4) 

21) ***Assertion (A) :-*** Benzene reacts with excess of Cl_2 in presence of sunlight to form BHC.
Reason (R) :- Reaction undergoes electrophilic substitution.

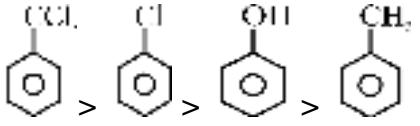
(1) Both **(A)** and **(R)** are correct but **(R)** is not the correct explanation of **(A)**

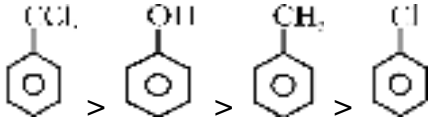
(2) **(A)** is correct but **(R)** is not correct

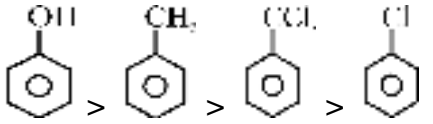
(3) **(A)** is incorrect but **(R)** is correct

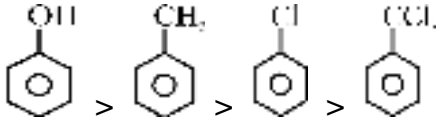
(4) Both **(A)** and **(R)** are correct but **(R)** is the correct explanation of **(A)**

22) Rate of Nitration (ESR) is :-

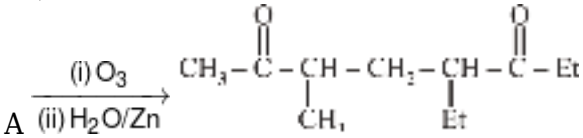
(1) 

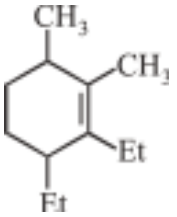
(2) 

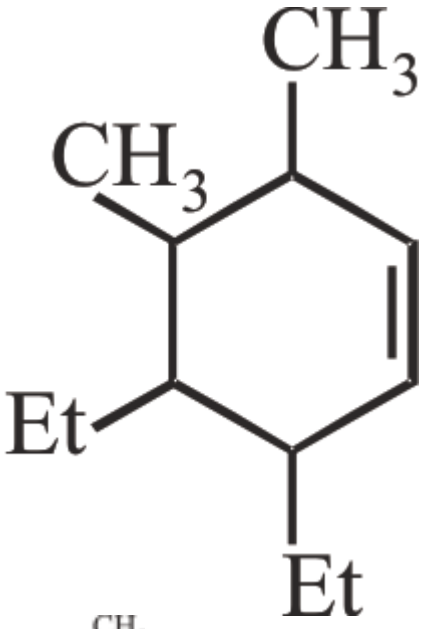
(3) 

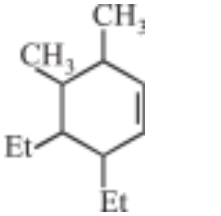
(4) 

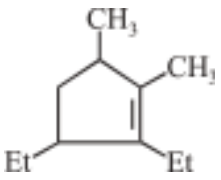
23) Which one is correct structure of "A"

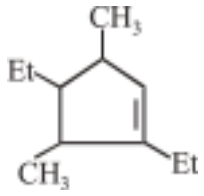
A 

(1) 

(2) 

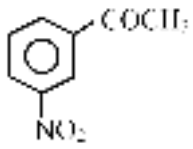
(3) 



(4) 

24) Which of the following compounds will not give white precipitate with Tollen's reagent-

(1) Ethyne
(2) 1-Butyne
(3) 3-Methyl-1-butyne
(4) 2-Pentyne

25) **Statement I :** Benzene is nitrated to give nitrobenzene which on further treatment with $CH_3COCl/AlCl_3$ will give following compound.  **Statement II :** $-NO_2$ group is a meta directing and deactivating group.

(1) Statement I is correct but Statment II is incorrect
(2) Both Statement are correct
(3) Statement I is incorrect but Statement II is correct
(4) Both Statement are incorrect

26) In a Lassigne's test for sulphur in the organic compound with sodium nitroprusside solution the purple colour formed is due to-

(1) $[Fe(CN)_5\ NOS]^{4-}$
(2) $[Fe(CN)_5\ S]^{2-}$
(3) $[Fe(CN)_5\ NOS]^{2-}$
(4) $[Fe(CN)_6\ ]^{4-}$

27) **Assertion (A) :–** A mixture of o–nitrophenol and p–nitrophenol can be separated by steam distillation.
Reason (R) :– o–nitrophenol is steam volatile but p–nitrophenol is not.

(1) Both (A) and (R) are correct and (R) is correct explanation of (A).
(2) Both (A) and (R) are correct but (R) is not correct explanation of (A).
(3) (A) is correct but (R) is wrong.
(4) Both (A) and (R) are wrong.

28) The volume occupied by 20g water at 1.2 atm and 4°C is about.

(1) $\frac{20 \times 0.082 \times 277}{18 \times 1.2}$L

(2) $\frac{20 \times 0.082 \times 4}{18 \times 1.2}$L

(3) 20 ml

(4) 17.2 L

29) The minimum mass of water needed to slake 1 kg of quicklime?

(1) 243.2 g

(2) 642.8 g

(3) 160.7 g

(4) 321.4 g

30) Find the value of K_C of reaction
$I_2(aq) + I^-(aq) \rightleftharpoons I_3^-(aq)$
If initial moles of I_2 and I^- are 1 mole and 0.5 mole respectively in 1 L flask. After equilibrium is reached, excess of $AgNO_3$ gave 0.25 mol of yellow precipitate

(1) 0.75

(2) 0.375

(3) 1.33

(4) 2.66

31) $SCN^-(aq) + Fe^{+3}(aq) \rightleftharpoons [Fe(SCN)^{+2}](aq)$
colourless yellow deep red
If thiocyanate ions are added in above equilibrium mixture.

(1) The solution becomes colourless.

(2) The yellow colour of solution deepens.

(3) Concentration of $[Fe(SCN)]^{+2}$ ion will decrease.

(4) The red colour of the solution deepens.

32) $2SO_2(g) + O_2(g) \rightleftharpoons 2SO_3(g)$, $\Delta H = -198$ kJ
Equilibrium constant of above reaction is changed by

(1) Adding Ne gas to reaction vessel at constant pressure

(2) Adding more of O_2 to the reaction vessel.

(3) Increasing the temperature.

(4) Doubling the volume of reaction vessel.

33) Find the value of K_P (in atm^2) for reaction.
$NH_4HS(s) \rightleftharpoons NH_3(g) + H_2S(g)$ at a certain temperature and total 3 atm pressure at equilibrium will be

(1) 2.25

(2) 1

(3) 4

(4) 9

34) Calculate pH of 0.02 M-HA solution.
[Given K_a of HA = 2×10^{-12}]

(1) 6.59
(2) 6.7
(3) 6.85
(4) None of these

35) Solubility of BaF_2 in a solution of $Ba(NO_3)_2$ will be represented by the concentration term :-

(1) $[Ba^{+2}]$
(2) $[F^-]$
(3) 0.5 $[F^-]$
(4) $2\left[NO_3^-\right]$

36) The solubility product of $Co(OH)_3$ is 2.7×10^{-43} . The pOH of saturated solution of $Co(OH)_3$ is about :

(1) 3.48
(2) 7.52
(3) 10.48
(4) 10.52

37) **Assertion (A) :-** Aqueous solution of NaCl and CH_3COONH_4 are neutral (pH = 7 at 25°C). (K_a & K_b are same for CH_3COOH and NH_4OH)
Reason (R) :- Both Salts do not undergo hydrolysis.

(1) Both **Assertion** and **Reason** are true and **Reason** is the correct explanation of **Assertion**.
(2) Both **Assertion** and **Reason** are true but **Reason** is NOT the correct explanation of **Assertion**.
(3) **Assertion** is correct but **Reason** is incorrect.
(4) **Assertion** is incorrect but **Reason** is correct.

38) For an isothermal process the essential condition is :

(1) $\Delta T = 0$
(2) $\Delta H = 0$
(3) $\Delta U = 0$
(4) $dT = 0$

39) Molar heat capacity of water in equilibrium with ice at constant pressure is :

(1) Zero
(2) 40.45 kJ/K-mol
(3) 75.48 J/K-mol

(4) Infinity

40) **Assertion (A) :-** Decrease in free energy causes spontaneous reaction.
Reason (R) :- Spontaneous reactions are invariably exothermic.

(1) Both **Assertion** and **Reason** are true and **Reason** is the correct explanation of **Assertion**.

(2) Both **Assertion** and **Reason** are true but **Reason** is NOT the correct explanation of **Assertion**.

(3) **Assertion** is correct but **Reason** is incorrect.

(4) **Assertion** is incorrect but **Reason** is correct.

41) Equivalent weight of H_2SO_4 in the reaction
$2KMnO_4 + 3H_2SO_4 + 10HCl \rightarrow$
$2MnSO_4 + K_2SO_4 + 5Cl_2 + 8H_2O$

(1) M

(2) $\frac{M}{2}$

(3) $\frac{3M}{10}$

(4) $\frac{3M}{5}$

42) Match the following :

Column-I (Process)		**Column-II (Equivalent weight of Cl_2)**	
(A)	$Cl_2 \rightarrow ClO_3^-$	(P)	35.5
(B)	$Cl_2 \rightarrow Cl^- + ClO^-$	(Q)	42.6
(C)	$Cl_2 \rightarrow Cl^-$	(R)	71
(D)	$Cl_2 \rightarrow Cl^- + ClO_3^-$	(S)	7.1

	A	**B**	**C**	**D**
(1)	P	Q	R	S
(2)	S	R	Q	P
(3)	S	Q	P	R
(4)	S	R	P	Q

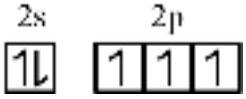
(1) 1

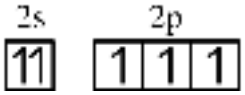
(2) 2

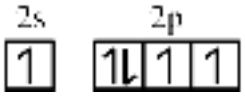
(3) 3

(4) 4

43) Which of the following configuration is violating Paul's exclusion principle?

(1) 

(2) 

(3) 

(4) (2) and (3) both

44) The number of orbitals in g-subshell :-

(1) 5
(2) 7
(3) 9
(4) 11

45) **Assertion (A) :-** For Li^{+2} ion the energy of electron in 3p-orbital is greater than that in 3s-orbital.
Reason (R) :- Energy of an electron in single electron system depends on n.

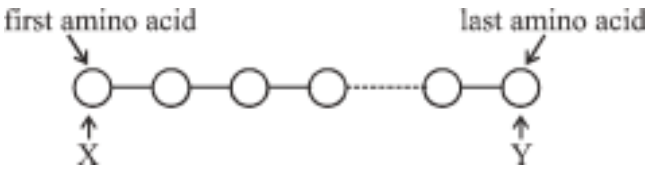
(1) Both **Assertion** and **Reason** are true and **Reason** is the correct explanation of **Assertion**.
(2) Both **Assertion** and **Reason** are true but **Reason** is NOT the correct explanation of **Assertion**.
(3) **Assertion** is correct but **Reason** is incorrect.
(4) **Assertion** is incorrect but **Reason** is correct.

BIOLOGY

1) The activity of an enzyme can be affected by a change in the conditions which can alter the :-

(1) Primary structure of the lipid
(2) Tertiary structure of the protein
(3) Primary structure of the carbohydrates
(4) Primary structure of fats

2) Identify X and Y in the given sequence :-



	X	Y
1	N-terminal amino acid	C-terminal amino acid
2	N-terminal amino acid	N-terminal amino acid
3	C-terminal amino acid	N-terminal amino acid
4	C-terminal amino acid	C-terminal amino acid

(1) 1

(2) 2

(3) 3

(4) 4

3) **Assertion (A) :** Enzymes isolated from thermophilic organisms can work efficiently even at 80-90^{0}C.
Reason (R) : These enzymes have high thermal stability and remain catalytically active at high temperatures.

(1) Both A and R are true and R is the correct explanation of A

(2) Both A and R are true but R is NOT the correct explanation of A.

(3) A is true but R is false

(4) A is false but R is true

4) Primary structure of protein is due to the presence of :-

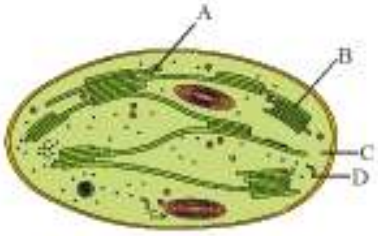
(1) Peptide Bond

(2) Hydrogen Bond

(3) Ionic Bond

(4) Disulphide Bond

5) In the given diagram of chloroplast identify the functions of the parts marked as A, B, C & D :-



	A	B	C	D
(1)	Light reaction	Only ATP synthesis	Dark reaction	Sugar storage
(2)	Dark reaction	Only ATP synthesis	Sugar synthesis	Sugar storage
(3)	Cyclic photophosporylation	Sugar synthesis	RuBisCO synthesis	Protein synthesis
(4)	Cyclic photophosphorylation only	Non cyclic photophosphorylation	Dark reaction	Protein synthesis

(1) 1

(2) 2

(3) 3

(4) 4

6) Number of carbons in the first CO_2 acceptor and first stable product in C_4 and C_3 cycle respectively are:-

(1) 3-C, 4C , 5C, 3C

(2) 5-C, 3-C , 2-C, 4-C

(3) 4-C, 3-C , 3-C, 4-C

(4) 2-C, 3-C , 4-C, 3-C

7) The reaction centre in PS II has an absorption maxima at

(1) 700 nm

(2) 660 nm

(3) 780 nm

(4) 680 nm

8) Which of the following are required for the dark reaction of photosynthesis?
(A) Light (B) Chlorophyll
(C) CO_2 (D) ATP
(E) NADPH
Choose the correct answer from the options given below:

(1) A, B and C only

(2) B, C and D only

(3) C, D and E only

(4) D and E only

9) Given below are two statements :
Statement I: RuBisCO is the most abundant enzyme in the world.
Statement II: Photorespiration does not occur in C_4 plants.
In the light of the above statements, choose the most appropriate answer from the options given below:

(1) Statement I is correct but Statement II is incorrect

(2) Statement I is incorrect but Statement II is correct

(3) Both Statement I and Statement Il are correct

(4) Both Statement I and Statement Il are incorrect

10) Enzyme found and attached to inner membrane of mitochondria instead of matrix is/are

(1) Malic dehydrogenase

(2) Succinic dehydrogenase

(3) Cytochrome c oxidase

(4) Both 2 and 3

11) Match the column I and column II.

	Column I		**Column II**

a.	2C compound	i.	Citric acid
b.	3C compound	ii.	Pyruvic acid
c.	4C compound	iii.	Malic acid
d.	5C compound	iv.	Acetyl CoA
		v.	α-Ketoglutaric acid

(1) a-iv, b-iii, c-ii, d-v

(2) a-iv, b-ii, c-iii, d-i

(3) a-iv, b-ii, c-iii, d-v

(4) a-ii, b-iii, c-iv, d-i

12) Choose the **correct** option for the products of alcoholic fermentation :-

(1) Alcohol, O_2 and NAD^+

(2) Alcohol + NADH + O_2

(3) Alcohol, CO_2 and NAD^+

(4) Alcohol, CO_2 + NADH

13) Complete the given reaction of the cycle present in mitochondrial matrix

A + Acetyl CoA + H_2O $\xrightarrow{B}$ Citric acid

(1) **A**-Succinic acid; **B**-Pyruvate dehydrogenase

(2) **A**-Malic acid; **B**-Transacetylase

(3) **A**-OAA; **B**-Citrate synthase

(4) **A**-α-KGA; **B**-Thiokinase

14) In link reaction from pyruvate to acetyl CoA formation is -

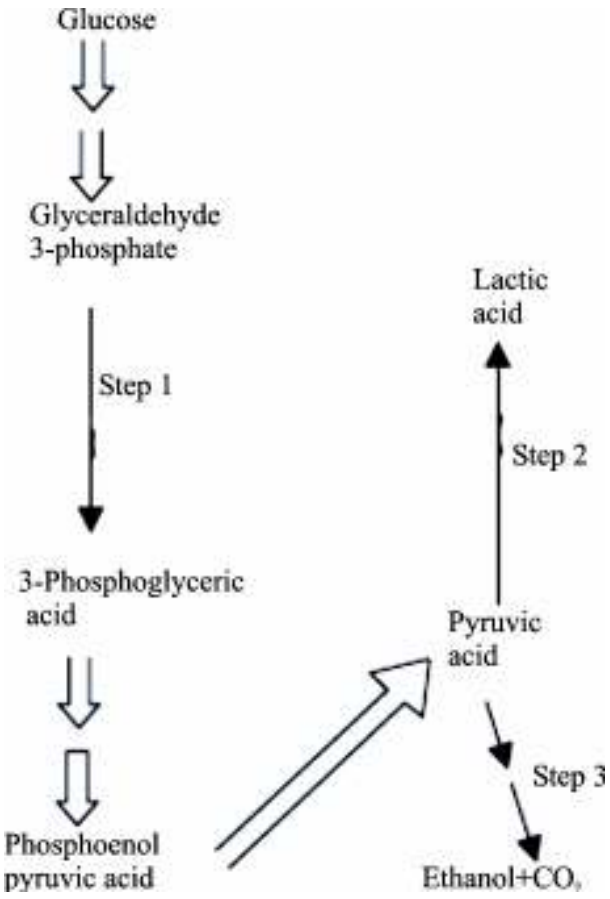
(1) Reductive decarboxylation

(2) Oxidative decarboxylation

(3) Substrate level phosphorylation

(4) Both 2 & 3

15) Pathways of anaerobic respiration are shown in the diagram. Select the correct option :-



	Synthesis of NADH + H^+	**Utilisation of NADH + H^+**
1	Step-2	Step-1 & 3
2	Step-1	Step-2 & 3
3	Step-2 & 3	Step-1
4	Step-1 & 3	Step-2

(1) 1

(2) 2

(3) 3

(4) 4

16)

	Column-I		**Column-II**
I	IAA	A	Terpenes
II	GA	B	Indole compounds
III	ABA	C	Adenine derivatives
IV	C_2H_4	D	Gases
V	Kinetin	E	Carotenoid derivatives

The correct match is

(1) 1-B, II-A, III-E, IV-D, V-C

(2) I-A, II-B, lll-C, IV-D, V-E

(3) I-E, ll-D, llI-A, IV-B, V-C

(4) I-C, ll-B, llI-A, IV-D

17) Select how many statements are correct from following ?
(A) Growth in plants is open.
(B) Plant growth is localised
(C) Decrease growth per unit time is termed as growth rate.
(D) Plant growth is affected by many factor

(1) One

(2) Two

(3) Three

(4) Four

18) Which of the following is correct :

(1) Auxin was isolated by F.W. Went from tips of coleoptiles of oat seedlings.

(2) 'Bakane' (foolish seedling) a disease of rice seedlings, was caused by a fungal pathogen *Gibberella fujikuroi*

(3) E. Kurosawa reported the appearance of symptoms of the disease in uninfected rice seedlings when they were treated with sterile filtrates of the fungus.

(4) All of these

19)

Match the column I with column II.

	Column I		**Column II**
a.	Extrinsic factors	i.	Temperature
b.	Intrinsic factors	ii.	Genetic factor
		iii.	Light
		iv.	Plant growth regulator

(1) a – i, iv b – ii, iii

(2) a – i, iii , iv b – ii

(3) a – i, iii b – ii, iv

(4) a – ii b – i, iii, iv

20) Study the following statements :-
(A) Bacteria are sole members of the kingdom monera
(B) Bacteria are most abundant micro-organisms
(C) A handful of soil contain thousands of bacteria
(D) In bacteria, mesosome help in respiration
(E) Bacteria show most extensive metabolic diversity

(1) A, B & E are correct

(2) A, C & E are correct

(3) A, B, D & E are correct

(4) A, B, C, D and E are correct

21) **Statement-I :-** In ascomycetes the asexual spores are conidia produced endogenously on the special mycelium called conidiophores.
Statement-II :- In ascomycetes sexual spores are called ascospores which are produced exogenously in sac like asci.

(1) Statement-I and II both are correct.

(2) Statement-I and II both are incorrect.

(3) Only statement-I is correct.

(4) Only statement-II is correct.

22) Match the column-I with column-II and choose the correct option :

	Column-I		**Column-II**
A.	Phycomycetes	(i)	Morels
B.	Ascomycetes	(ii)	*Alternaria*
C.	Basidiomycetes	(iii)	*Mucor*
D.	Deuteromycetes	(iv)	*Ustilago*

(1) A-i, B-iv, C-iii, D-ii

(2) A-iii, B-i, C-iv, D-ii

(3) A-ii, B-i, C-i, D-iv

(4) A-iii, B-ii, C-iv, D-i

23) Select the incorrect statement from following -

(1) The earliest system of classification used only gross superficial morphological character such as habit, colour, number and shape of leaves etc.

(2) Structure of androecium was used by Linnaeus for classification of plants.

(3) System of classification given by Linnaeus was artificial.

(4) Bentham and Hooker proposed phylogenic classification system.

24) Match the **Column-I** & **Column-II** and select correct option.

Column I		**Column II**	
A	Diatoms	i.	Pigments identical to higher plants
B	Euglenoids	ii.	Primitive relatives of animals
C	Slime moulds	iii.	Soap box like cell wall
D	Protozoans	iv.	Saprophytic protist

(1) A-i, B-ii, C-iii, D-iv

(2) A-iii, B-i, C-iv, D-ii

(3) A-ii, B-iii, C-i, D-iv

(4) A-iv, B-iii, C-ii, D-i

25) Given organisms belongs to how many families ?
Wheat, Mango, Brinjal, Potato, Datura, Lion, Cat, Dog

(1) Seven

(2) Six

(3) Five

(4) Four

26) Select the correct statement from the following:-
(a) Family has a group of related orders.
(b) Cats and dogs belong to the same family called canidae.
(c) *Panthera* and *felis* belongs to the family felidae.
(d) Families are characterised only on the basis of reproductive features of plant species.

(1) a, b and d only

(2) only d

(3) only c

(4) c and d only

27) Study the following statements carefully:-
(A) The kingdoms defined by R.H. Whittaker were named Monera, Protista, Fungi, Plantae and Animalia.
(B) The three-domain system divides the kingdom Monera into two domain, leaving the remaining eukaryotic kingdoms in the third domain.
(C) According to five kingdom classification *Chlamydomonas* and *Chlorella* both are placed in Algae.
(D) Domain is the super division category.
How many are correct statement?

(1) Four

(2) One

(3) Two

(4) Three

28) **Assertion (A) :** In *Volvox* and *Fucus* oogamous type of sexual reproduction is found.
Reason (R) : In *Volvox* and *Fucus* fusion between one large, non motile (static) female gamete and a smaller motile male gamete occurs.

(1) Both (A) and (R) are correct and (R) is the correct explanation of (A)

(2) (A) is correct but (R) is incorrect

(3) (A) is incorrect but (R) is correct

(4) Both (A) and (R) are correct but (R) is not the correct explanation of (A)

29) Which of the following is method of asexual reproduction in liverworts?

(1) Fragmentation of thalli

(2) Gemma formation

(3) Budding in secondary Protonema

(4) 1 and 2 both

30) Read all statements carefully regarding pteridophyte and select incorrect statement :

(1) Pteridophytes are used for medicinal purposes and soil-binders.

(2) All pteridophytes exhibits haplo-diplontic life cycle pattern.

(3) Precursor to the seed habit considered an important step in evolution in homosporous pteridophyte

(4) Pteridophytes are first vascular plant.

31) Match the column I with column II and choose the correct option -

	Column-I		**Column-II**
A	*Anabaena*	i	Independent gametophyte
B	*Cycas*	ii	Dioecious sporophyte
C	*Funaria*	iii	Pyriform gametes
D	*Ectocarpus*	iv	Nitrogen fixation

Option :

(1) A - i, B - iii, C - iv, D - ii

(2) A - iii, B - i, C - ii, D - iv

(3) A - iv, B - i, C - ii, D - iii

(4) A - iv, B - ii, C - i, D - iii

32) **Assertion (A) :-** Heterosporous pteridophytes show the event precursor to seed habit.
Reason (R) :- In heterosparous pteridophytes the female gametophytes are retained on the parent sporophytes for variable periods. The development of the zygotes into young embryos take place within the female gametophytes.

(1) Both (A) and (R) are correct and (R) is the correct explanation of (A)

(2) (A) is correct but (R) is incorrect

(3) (A) is incorrect but (R) is correct

(4) Both (A) and (R) are correct but (R) is not the correct explanation of (A)

33) Below two statement are given.
Statement I : The gymnosperm are plant in which the ovules are not enclosed by any ovary wall and remain exposed both before and after fertilisation.
Statement II : Gymnosperm include medium-sized trees or tall trees and shrubs.

(1) Both statement I and statement II are incorrect.

(2) Statement I is correct but statement II is incorrect.

(3) Statement I is incorrect but statement II is correct.

(4) Both statement I and statement II are correct.

34) Given below are the stage in the life cycle of pteridophytes arrange the following stages in correct sequence -
(A) Germination of spores
(B) Formation of embryo
(C) Fusion of male gamete with egg
(D) Formation of sporangia on sporophylls
(E) Sporophytic plant body bearing root, stem and leaves
Choose the correct answer from the options given below :

(1) B, E, D, C, A

(2) D, A, C, B, E

(3) D, C, A, E, B

(4) C, B, D, E, A

35) Meiosis ensures the production of _____ phase in the life cycle of sexually reproducing organisms, whereas fertilisation restores the _____ phase.

(1) haploid ; diploid

(2) haploid; haploid

(3) diploid ; haploid

(4) diploid ; diploid

36) If a drug disrupts the polymerization of tubulin, what specific component of the mitotic apparatus will be most directly affect?

(1) Asters only

(2) Chromosomes

(3) Spindle fibres

(4) Chromatids

37) **Statement-I :** In a mitotic cell cycle (eg. onion root tip cells) chromosome number is 16 in all the phases i.e., G_1 phase, s-phase and G_2 phase.
Statement-II : DNA content of the cells at G_1 is 2C, after S- and in G_2 is 4C, if the content after M-phase is 2C.

(1) Both **Statement I** and **Statement II** are incorrect.

(2) **Statement I** is correct but **Statement II** is incorrect.

(3) **Statement I** is incorrect but **Statement II** is correct.

(4) Both **Statement I** and **Statement II** are correct.

38) Cells having 20 bivalents undergoes meiosis I then calculate the number of chromatids in each nuclei after meiosis I.

(1) 20

(2) 40

(3) 10

(4) 80

39) Fill up the blanks in the following statements with the appropriate option :
(I) At _A_ the chromosome align at the equator.
(II) Meiosis increases the _B_ in the population of organism from one generation to the next.
(III) Meiosis II is equational division due to _C_ .
(IV) _D_ are formed after the cytokinesis-II.

	A	B	C	D
(1)	Metaphase-II	Genetic variability	Separation of chromatids	Tetrad of cells
(2)	Anaphase-II	Variations	Separation of chromosomes	Dyad of cells
(3)	Telophase-II	Evolution	Separation of chromatids	Tetrad of cells
(4)	Prophase-II	Genetic stability	Separation of Homologous chromosomes	Two Diploid cells

(1) 1

(2) 2

(3) 3

(4) 4

40) Primary wall gradually ____A____ as cell matures and secondary wall is formed on ____B____ side at the cell.

(1) A-diminish, B-Outer

(2) A-diminish, B-Inner

(3) A-Grow, B-Outer

(4) A-Grow, B-Inner

41) Match the following lists.

	List-I		List-II
(a)	Aleuroplast	(i)	Store oil and fats
(b)	Elaioplast	(ii)	Store proteins
(c)	Amyloplast	(iii)	Xanthophyll pigment
(d)	Chromoplast	(iv)	Store carbohydrates

	(a)	(b)	(c)	(d)

1	(i)	(ii)	(iii)	(iv)
2	(ii)	(i)	(iv)	(iii)
3	(iv)	(iii)	(ii)	(i)
4	(iv)	(iii)	(i)	(ii)

(1) 1

(2) 2

(3) 3

(4) 4

42) How many of the following statement are correct ?
(a) Endomembrane system includes endoplasmic reticulum (ER), golgi complex, lysosomes and vacuoles.
(b) Functions of mitochondria, chloroplast and peroxisomes are co-ordinated with endomembrane system.
(c) ER divides the intracellular space into two distinct compartments.
(d) In the absence of ribosomes ER appear smooth and are called smooth endoplasmic reticulum.

(1) four

(2) Three

(3) two

(4) One

43) Consider the following statements and choose the correct option.
(a) RER is major site for glycolipid synthesis
(b) Mitochondria, chloroplast and peroxisomes are not part of endomembrane system.
(c) The stroma of the chloroplast contains enzymes required for the synthesis of carbohydrates and proteins.
(d) F1 particles occur in ribosomes.

(1) (a), (b), (c) and true

(2) (a), (c), (d) are true

(3) (b), (c) are true

(4) (a), (d) are true

44) The vacuole is the membrane-bound space found in the cytoplasm. It contains-
A. Water
B. Sap
C. Excretory compounds
D. Other undersirable substances

(1) Only A and B

(2) Only C and D

(3) Only A, B and D

(4) All A,B, C and D

45) **Assertion (A) :** Lysosomes are more active in Phagocytic cells.

Reason (R) : Phagocytic cells perform intracellular digestion using lysosomal enzymes.

(1) Both (A) and (R) are true and (R) is the correct explanation of (A).

(2) Both (A) and (R) are true but (R) is not the correct explanation of (A).

(3) (A) is true but (R) is false.

(4) (A) is false but (R) is true.

46) Match the column-I with column-II and choose correct option :

Column-I		**Column-II**	
a	Chordata	i	Agnatha, jawless Vertebrates
b	Cephalochordata	ii	Presence of notochord throughout life
c	Cyclostomata	iii	Dorsal hollow nerve cord
		iv	Amphioxus
		v	Lamprey

(1) a-iii, b-ii, c-i

(2) a-ii, b-v, c-iv

(3) a-iv, b-v, c-i

(4) a-i, b-v, c-iv

47) Match the column-I with column-II on the basis of phylum and example and choose the correct option :

Column-I (Phylum)		**Column-II (Example)**	
A	Porifera	i	Pleurobrachia
B	Cnidaria	ii	Taenia
C	Ctenophora	iii	Spongilla
D	Platyhelminthes	iv	Aurelia

(1) A-iii, B-iv, C-i, D-ii

(2) A-i, B-iii, C-ii, D-iv

(3) A-iv, B-ii, C-i, D-iii

(4) A-ii, B-i, C-iv, D-iii

48) In which vertebrates mouth is located ventrally and notochord is persistent throughout life :-

(1) Urochordata

(2) Chondricthyes

(3) Osteichthyes

(4) Cephalochordata

49) An animal is poikilothermous but it's not a fish, It has three chambered heart but not an amphibian. It is oviparous but not a bird this animal may be -

(1) Exocoetus

(2) Crocodilus

(3) Rana

(4) Chameleon

50) **Assertion (A) :-** Osteichthyes fishes do not have to swim constantly to avoid sinking.
Reason (R) :- They possess an air bladder which regulates buoyancy.

(1) Both A and R are true and R is the correct explanation of A

(2) Both A and R are true but R is not the correct explanation of A

(3) A is true but R is false

(4) A is false but R is true

51) Which of the following animal is pseudocoelomate ?

(1) Pheretima

(2) Ascaris

(3) Canis

(4) Echinus

52) The main function of simple cuboidal epithelium is _____ & _____

(1) Filtration & diffusion

(2) Secretion & absorption

(3) filtration & absorption

(4) secretion & diffusion.

53) In cockroach touch, temperature and vibration receptor are present. These sensory receptors present on ..... help in monitoring the environment.
Fill in the blank :

(1) Mandibles

(2) Antennae

(3) Thorax

(4) Compound eye

54) Given below are two statements ;
Statement-I : Male reproductive system of cockroach consists of 1 pair of testis located on lateral side from 2nd to 6th abdominal segments.
Statement-II : Each testis formed of 3 or 4 lobes and each lobe divided into many lobules.

In the light of the above statements choose the most appropriate answer from the options given below :

(1) Both statement I and statement II are correct

(2) Both statement I and statement II are incorrect

(3) Statement I is correct but statement II is incorrect

(4) Statement I is incorrect but statement II is correct

55) Which one of the following statements is correct about Rana tigrina (Adult) ?

(1) During aestivation and hibernation, gaseous exchange takes place through lungs.

(2) In water, gills act as aquatic respiratory organ.

(3) The body is divisible into the head, neck and trunk.

(4) Fertilisation is external and development is indirect.

56) Which one of the following is common in both male and female frog?

(1) Bidder's canal

(2) Vocal sac

(3) Cloaca

(4) Copulatory pad

57) Why are the pulmonary veins an exception to other veins ?

(1) It carries deoxygenated blood

(2) It opens into the right atrium

(3) It carries oxygenated blood

(4) It carries the blood from kidney

58) During which stage tricuspid and bicuspid valve remains open ?

(1) Joint diastole

(2) Atrial Diastole

(3) Ventricular systole

(4) 1 and 2 both

59) **Statement-I :-** During a cardiac cycle, each ventricle pumps out approximately 70 mL of blood which is called the stroke volume. The stroke volume multiplied by the heart rate gives the cardiac output.
Statement-II :- During each cardiac cycle two prominent sounds are produced which can be easily heard through a stethoscope

(1) Both Statement-I and Statement-II are correct.

(2) Both Statement-I and Statement-II are incorrect.

(3) Statement-I is correct but Statement-II is incorrect.

(4) Statement-I is incorrect but Statement-II is correct.



60) Identify the following diagram and select the correct option?

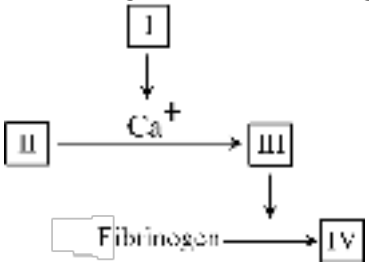
(1) Eosinophil
(2) Basophil
(3) Monocyte
(4) T-lymphocyte

61) Match the column-I with column-II and III

Column-I		Column-II		Column-III	
I		a	0.5-1%	W	Histamin secretes
II		b	60-65%	X	Antibody formation
III		c	20-25%	Y	Phagocytic
IV		d	6-8%	Z	Monocyte

(1) I-b-Y, II-a-W, III-d-Z, IV-c-X
(2) I-a-Y, II-b-W, III-d-Z, IV-c-X
(3) I-a-W, II-b-X, III-c-Y, IV-d-Z
(4) I-d-Z, II-c-Y, III-b-X, IV-a-W

62) Recognise the following below diagram and find correct option according to blood coagulation?



(1) I-prothrombin, II-thrombin, III-thrombokinase, IV-clot

(2) I-thrombokinase, II-prothrombin, III-thrombin, IV-fibrins

(3) I-thromboplastin, II-prothrombin, III-thrombin, IV-clot

(4) I-platelets, II-thrombin, III-prothrombin, IV-fibrins.

63) In humans the primary site for exchange of gases is –

(1) Brohchi

(2) Trachea

(3) Brohchioles

(4) Alveoli

64) **Statement I :** Human being have significant ability to maintain and moderate the respiratory rhythm to suit the demands of body tissues.
Statement II : Alveoli are made up of simple squamous epithelium.

(1) Both **Statement I** and **Statement II** are correct

(2) Both **Statement I** and **Statement II** are incorrect

(3) **Statement I** is correct but **Statement II** is incorrect

(4) **Statement I** is incorrect but **Statement II** is correct.

65) Match the **Column I** with **Column II** and select the correct option.

	Column I		**Column II**
A	Vital capacity	(i)	TV + ERV
B	Expiratory capacity	(ii)	TV + IRV
C	Inspiratory capacity	(iii)	ERV + RV
D	Functional Residual capacity	(iv)	TV + ERV + IRV

(1) A-IV, B-I, C-III, D-II

(2) A-I, B-II, C-III, D-IV

(3) A-II, B-I, C-III, D-IV

(4) A-IV, B-I, C-II, D-III

66) In the alveoli, oxyhaemoglobin formation occurs, factors which favour formation of oxyhaemoglobin includes.

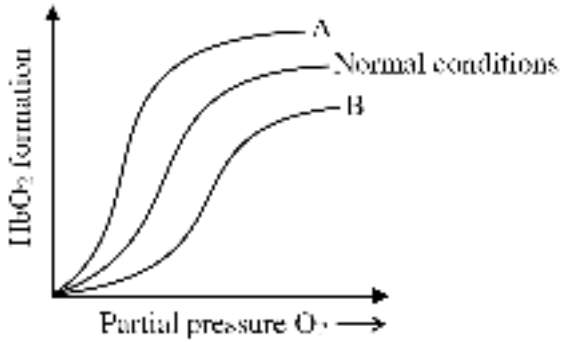
(1) high p^{O_2} , low p^{CO_2} , low H^+, low Temperature

(2) high p^{O_2} , high p^{CO_2} , low H^+, high Temperature

(3) low p^{O_2}, high p^{CO_2}, high H^+, high temperature

(4) low p^{O_2}, low p^{CO_2}, low H^+, low temperature

67) Observe the given diagram of oxygen dissociation curve and choose the incorrect about it :-



(1) B → High H^+ concentration

(2) B → High temperature

(3) B → Low P_{CO_2}

(4) A → High P_{O_2}

68) Given below are two statements.
Statement-I: PCT is lined by simple cuboidal brush border epithelium which increases the surface area for reabsorption. PCT helps to maintain the pH and ionic balance of the body fluids by selective secretion of hydrogen ions and HCO_3^-.
Statement-II: Conditional reabsorption of Na^+ and K^+ takes place from DCT. It is also capable of reabsorption of HCO_3^- and selective secretion of hydrogen ions, chloride ions and NH_3 to maintain the pH and ion balance in blood.

(1) Both statement-I and II are correct.

(2) Both statement-I and II are incorrect.

(3) Statement-I is incorrect but II is correct.

(4) Statement-I is correct but II is incorrect.

69) **Assertion: -** The Juxta medullary nephrons produces concentrated urine.
Reason: In some nephrons, the loop of henle is very long and runs deep into the medulla. These nephrons are called cortical nephrons.

(1) **Assertion** and **Reason** both are true and **Reason** is correct explanation of **Assertion**

(2) **Assertion** and **Reason** both are true but remains is not correct explanation of **Assertion**

(3) **Assertion** is true but **Reason** is false

(4) **Assertion** and **Reason** both are false.

70) Select the correct statements -

(1) Angiotensin II is a powerful vasoconstrictor.

(2) Atrial natriuretic factor increases the blood pressure.

(3) Counter current pattern of blood flow is not observed in vasa recta.

(4) Increase in GFR activates JG cells to release remain.

71) Which part of Nephron helps to maintain concentration gradiente of medullar interstitium ?

(1) PCT

(2) Loop of Henle

(3) DCT

(4) Bowman's Capsule

72) What amount of CO_2 are removed by lungs in 6 hrs ?

(1) 1200 ml

(2) 12000 ml

(3) 72000 ml

(4) 7200 ml

73)

Select the option which appropriately. Fills (i), (ii) & (iii) :

Neuron Type	**Description**
Multipolar	(i)
Bipolar	(ii)
Unipolar	(iii)

(1) (i) One dendrite only, (ii) many dendrites,
(iii) embryonic stage only

(2) (i) 1 axon+ many dendrites,
(ii) 1 axon+ 1 dendrite, (iii) 1 axon only

(3) (i) No dendrites,
(ii) 2 axons, (iii) 2 dendrites

(4) (i) Mostly sensory,
(ii) Only motor, (iii) Only CNS

74) When a neuron is not conducting any impulse i.e. resting, the axonal membrane is -

(1) Comparatively more permeable to K^+ and impermeable (nearly impermeable) to Na^+

(2) Impermeable to negatively charged proteins present in the axoplasm.

(3) (1) & (2) Both

(4) More permeable to Na^+ ions than K^+ ion.

75) (A) Limbic system along with hypothalamus is involved in expression of emotional reactions.
(B) Dorsal portion of midbrain consist mainly of four round swellings.
(C) A canal called cerebral aqueduct passes through medulla oblongata.
(D) Midbrain, cerebellum, Medulla oblongata makeup the brain stem.
(E) Hypothalamus lies at the base of thalamus.
How many of the given statements are correct?

(1) A, B, D & E

(2) B, C, D & E

(3) A, B & E

(4) A, C, D & E

76) Ion channels are _____ for different ions :

(1) Semi permeable

(2) Permeable

(3) Selectively permeable

(4) Impermeable

77) Match List-I with List-II :-

	List-I		**List-II**
(A)	Limbic system	(I)	Tract of nerves
(B)	Dura matter	(II)	Urge for eating and drinking
(C)	Corpus callosum	(III)	Cranial meninges
(D)	Hypothalamus	(IV)	Hippocampus

Choose the correct answer from the options given :-

(1) A-III, B-II, C-I, D-IV

(2) A-IV, B-I, C-II, D-III

(3) A-IV, B-III, C-I, D-II

(4) A-IV, B-III, C-II, D-I

78) Match the function with related part in this given table.

Part	**Function**
Adrenal medulla	Fight or flight response
.....A.....	Sleep wake cycle
.....B.....	Resorption of bone
Posterior pituitary	.....C.....

(1) A–Pineal gland, B–Parathyroid, C–Increases BMR

(2) A–Thymus gland, B–Parathyroid, C–Parturition

(3) A–Pituitary, B–Thyroid, C–Milk secretion

(4) A–Pineal gland, B–Parathyroid, C–Parturition

79) Match column-I with column-II.

	Column-I		**Column-II**

(A)	Secreted by Posterior pituitary	(i)	Prolactin
(B)	Secreted by Parathyroid gland	(ii)	ACTH
(C)	Stimulates Adrenal gland	(iii)	Thyrocalcitonin
(D)	Secreted by Anterior pituitary	(iv)	PTH
(E)	Secreted by Thyroid gland	(v)	ADH

(1) (A)→v ; (B)→iv ; (C)→ii ; (D)→iii ; (E)→i

(2) (A)→v ; (B)→iv ; (C)→i ; (D)→ii ; (E)→iii

(3) (A)→iv ; (B)→v ; (C)→ii ; (D)→i ; (E)→iii

(4) (A)→v ; (B)→iv ; (C)→ii ; (D)→i ; (E)→iii

80) Select the incorrect statement -

(1) Insulin is a hypoglycemic hormone.

(2) Glucagon is a peptide hormone that stimulates glycogenesis.

(3) Androgen produce anabolic effects on protein and carbohydrate metabolism.

(4) Progesterone stimulates alveolar development of mammary gland.

81) Which gland secretes a hormone that helps increase blood calcium levels ?

(1) Thyroid

(2) Pituitary

(3) Parathyroid

(4) Pineal

82) Addison's disease is caused due to :-

(1) Hypersecretion of adrenal cortex hormone.

(2) Hyposecretion of adrenal cortex hormone.

(3) Hypersecretion of thyroid gland hormone.

(4) Hyposecretion of adrenal medulla hormone.

83) The main function of melatonin hormone is to control :

(1) Metabolism

(2) Pigmentation

(3) Menstrual cycle

(4) Diurnal rhythm

84) Select the incorrect statement given below :

(1) The skull region articulates with the superior region of the vertebral column with help of two occipital condyle.

(2) Vertebral column extends from the base of the skull and constitute the main framework of the trunk

(3) Sternum is a flat bone on the dorsal midline of thorax.

(4) There are 12 pair of ribs, each rib is a thin flat bone connected dorsally to the vertebral column and ventrally to the sternum directly.

85) **Statement-I :** Osteoporosis is a degenerative joint disease characterized by the degeneration of the articular cartilage and proliferation of new bones.
Statement-II : In gout disorder inflammation of the joints occurs due to accumulation of uric acid crystals.
Select the correct option from the following :

(1) Both statement-I and statement-II are correct.

(2) Statement-I is correct and statement-II is incorrect.

(3) Statement-I is incorrect and statement-II is correct.

(4) Both statement-I and statement-II are incorrect.

86) Match the following and mark the correct option :

	Column-I		**Column-II**
(A)	Cartilaginous joint	(i)	Between the carpals
(B)	Gliding joint	(ii)	Between carpals and metacarpals of thumb
(C)	Fibrous joint	(iii)	Pubic symphysis
(D)	Saddle joint	(iv)	Between the cranial bones

(1) A→iii, B→ii, C→iv, D→iii

(2) A→iii, B→i, C→iv, D→ii

(3) A→ii, B→i, C→iv, D→iii

(4) A→iii, B→iv, C→ii, D→iii

87) Which is triangular flat bone of the body?

(1) Scapula bone

(2) Sternum bone

(3) Sphenoid bone

(4) Collar bone

88) Fill in the blanks :-
Each coxal bone is formed by the fusion of three bones ___A___. At the point of furion of the above bones is a cavity called___B___.

(1) A - Ilium, ischium, pubis,
B - Glenoid.

(2) A - Ilium, ischium, pubis,
B - Foramen magnum

(3) A - Ilium, ischium, pubis,
B - Acetabulum

(4) A - Ilium, ischium, patella,
B - Acetabulum

89) Myosin binding site is present on :

(1) Troponin
(2) Tropomyosin
(3) G-actin
(4) Meromyosin

90) Which part of meromyosin protein not include in heavy meromyosin?

(1) Short arm
(2) Globular head
(3) 1 and 2 both
(4) Tail part

ANSWER KEYS

PHYSICS

Q.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
A.	2	4	4	2	3	3	4	2	4	2	2	3	2	4	1	3	3	3	2	3
Q.	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40
A.	1	2	4	3	4	1	1	2	3	1	2	1	1	1	2	1	1	1	1	2
Q.	41	42	43	44	45															
A.	1	1	2	2	3															

CHEMISTRY

Q.	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60	61	62	63	64	65
A.	1	2	2	2	4	3	1	2	1	2	3	1	4	3	2	3	4	3	3	4
Q.	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85
A.	2	4	3	4	3	1	1	3	4	3	4	3	1	1	3	4	3	4	4	3
Q.	86	87	88	89	90															
A.	3	4	2	3	4															

BIOLOGY

Q.	91	92	93	94	95	96	97	98	99	100	101	102	103	104	105	106	107	108	109	110
A.	2	1	1	1	4	1	4	3	3	4	3	3	3	2	2	1	3	4	3	4
Q.	111	112	113	114	115	116	117	118	119	120	121	122	123	124	125	126	127	128	129	130
A.	2	2	4	2	3	3	3	1	4	3	4	1	4	2	1	3	4	2	1	2
Q.	131	132	133	134	135	136	137	138	139	140	141	142	143	144	145	146	147	148	149	150
A.	2	2	3	4	1	1	1	2	4	1	2	2	2	4	4	3	3	1	1	1
Q.	151	152	153	154	155	156	157	158	159	160	161	162	163	164	165	166	167	168	169	170
A.	1	2	4	1	4	1	3	2	3	1	2	3	2	3	3	3	3	4	4	2
Q.	171	172	173	174	175	176	177	178	179	180										
A.	3	2	4	3	3	2	1	3	3	4										

SOLUTIONS

PHYSICS

1) $\tan\beta = \frac{v_y}{v_x}$

$\tan\beta = \frac{u_y - gt}{u_x}$

$\tan\beta = \tan\alpha - 5\sec\alpha$

2) NCERT Pg. # 51
Relative velocity

3) This equation is similar to the equation of trajectory of a projectile thrown horizontally from the top of the tower.

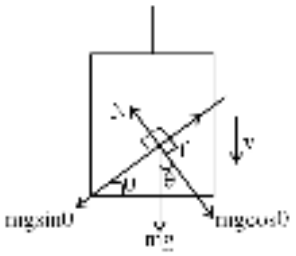
$y = \alpha x^2, y = \frac{gx^2}{2u^2} = \frac{\beta x^2}{2u^2}$

$\alpha = \frac{\beta}{2u^2} \Rightarrow u = \sqrt{\frac{\beta}{2\alpha}}$

4) $\vec{V}_{AB} = (2-x)\hat{i} + (3-y)\hat{j}$

$S_{rel} = u_{rel} \times t$

In y direction	In x direction
$0 = (3-y)\,2$	$2 = 2\,(2-x)$
$0 = 6 - 2y$	$2 - x = 1$
$2y = 6$	$x = 1$
$y = 3$	

5) 

Block is at rest so

$f = mg\sin\theta$

$W_f = |f|\,|s|\cos(90 + \theta)$

$W_f = (mg\sin\theta)(vt)(-\sin\theta)$

$W_f = -mg\sin^2\theta\,(vt)$ [s = vt]

$W_f = -mg\,vt\sin^2\theta$

6) Instantaneous power, $P = \vec{F} \cdot \vec{v}$

$\Rightarrow P = \left(10\hat{i} + 10\hat{j} + 20\hat{k}\right) \cdot \left(5\hat{i} - 3\hat{j} + 6\hat{k}\right)$

P = (10) (5) + (10) (–3) + (20) (6)

P = 50 – 30 + 120 = 140W = 140 joule/sec

7) In region AB and CD, slope of the graph is constant i.e. velocity is constant. It means no force acting on the particle in this region.

8) $F = u\left(\frac{dm}{dt}\right) = 500 \times 50 \times 10^{-3} = 25N$

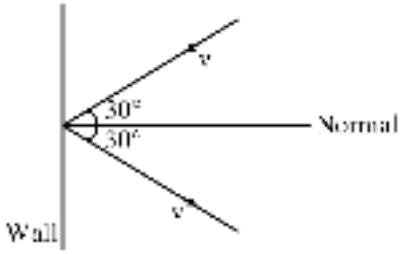
9) $\because x_{CM} = \frac{mL + 2m \times 2L + ... + nm \cdot nL}{m + 2m + ... + nm}$

$= \frac{L\left(1^2 + 2^2 + ... + n^2\right)}{(1 + 2 + ... + n)}$

$= L\frac{\sum n^2}{\sum n} = L\frac{\frac{n(n+1)(2n+1)}{6}}{\frac{n(n+1)}{2}}$

$= \frac{(2n+1)L}{3}$

10) Linear momentum remains conserved in elastic collision as well as in inelastic collision but kinetic energy remains conserved only in an elastic collision.



Given that the speed of the molecule of gas is same before and after the collision, therefore its kinetic energy is conserved. Hence the collision is elastic.

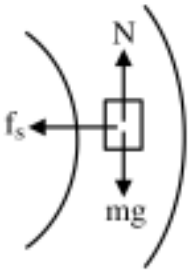
11) Apply COME from B to C :-

$\Rightarrow \frac{1}{2}m\left(\frac{1}{4}gr\right) + mg(r - r\cos\theta) = \frac{1}{2}mv^2$

$\Rightarrow v^2 = \frac{gr + 8gr(1 - \cos\theta)}{4}$ ...(I)

At C :- $mg\cos\theta = \frac{mv^2}{r}$ ...(II)

Put value of v^2 from (I) in (II) and find $\cos\theta$.



12)

$$f_s = \frac{mv^2}{R}$$

$$\mu N = \frac{mv^2}{R}$$

$$\mu mg = \frac{mv^2}{R}$$

$$v = \sqrt{\mu Rg}$$

$$v = \sqrt{\frac{1}{2} \times 45 \times 10} = 15 m/s$$

$$v = 15 \times \frac{18}{5} = 54 \text{ km/hr}$$

13) Here, Q and ΔU are same so from FLOT

Q = W + ΔU, we have $W_1 = W_2$

But W = PΔV So $P\Delta V_1 = P\Delta V_2$

$$\Rightarrow \pi r_A^2 x_A = \pi r_B^2 x_B$$

$$\Rightarrow \frac{r_A}{r_B} = \sqrt{\frac{x_B}{x_A}} = \sqrt{\frac{9}{16}} = \frac{3}{4}$$

14) $PV = \frac{N}{N_A} RT$

$PV = NKT \Rightarrow N = \frac{PV}{KT}$

$\therefore \frac{N_1}{N_2} = 4 : 1$

15)

$\therefore f_1 = f_2$

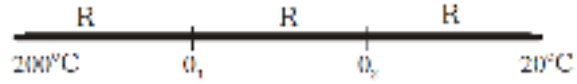
$$T_{mix} = \frac{n_1 T_1 + n_2 T_2}{n_1 + n_2}$$

$$= \frac{2T_0 + 4(2T_0)}{6}$$

$$= \frac{10T_o}{6} = \frac{5T_o}{3}$$



16) Resistance of all identical rods are same.



$H_1 = H_2 = H_3$

$$\frac{KA(200-\theta_1)}{\ell} = \frac{KA(\theta_1-\theta_2)}{\ell} = \frac{KA(\theta_2-20)}{\ell}$$

$\theta_1 = 140°C$ $\theta_2 = 80°C$

17) $W = P \times \Delta V$

(For isochoric processes, W = 0)

AB (Isobaric, volume decreasing)

$W_{AB} = P_0 \times (V_0 - 2V_0) = -P_0V_0$

BC (isochoric)

CD (Isobaric, volume increasing) :

$W_{CD} = 2P_0 \times (3V_0 - V_0) = 4P_0V_0$

Net work done :

$W_{ABCD} = W_{AB} + W_{CD}$

$W_{ABCD} = -P_0V_0 + 4P_0V_0 = 3P_0V_0$

18) From given curve,

Melting point for A = 60°C

and melting point for B = 20°C

Time taken by A for fusion = (6 – 2) = 4 minute

Time taken by B for fusion = (6.5 – 4) = 2.5 minute

Then, $\frac{H_A}{H_B} = \frac{6 \times 4 \times 60}{6 \times 2.5 \times 60} = \frac{8}{5}$

19) At $x = \frac{A}{4}$

$$K.E. = \frac{1}{2}K\left(A^2 - \frac{A^2}{4}\right) = \frac{1}{2}KA^2\left(\frac{15}{16}\right)$$

$$\frac{K.E.}{T.E.} = \frac{\frac{1}{2}KA^2\left(\frac{15}{16}\right)}{\frac{1}{2}KA^2} = \frac{15}{16}$$

20)

$V = \omega\sqrt{A^2 - x^2}$

$V^2 = \omega^2A^2 - \omega^2x^2$

$$\frac{V^2}{\omega^2A^2} + \frac{x^2}{A^2} = 1$$

graph is ellipse.

21) $T = 2\pi\sqrt{\frac{m}{k}}$

$T' = 2\pi\sqrt{\frac{m}{nk}} = \frac{T}{\sqrt{n}}$

22)

$I \propto n^2A^2$

$\frac{I_1}{I_2} = \left(\frac{n_1}{n_2}\right)^2\left(\frac{A_1}{A_2}\right)^2$

23)

$\Delta x = n\lambda$

$\Rightarrow (EA + AB + BF) - EF = n\lambda$

For minimum $\square$, n = 1

$1.2\square = 6 \Rightarrow \square = 5$ m

24) 1) $\Delta x = \frac{\lambda}{2}$ (Destructive)

$I_{net} = I_{min} = 0$

2) $\Delta x = 3\lambda$ (constructive)

$I_{net} = I_{max} = 4I$

3) $\Delta x = \frac{\lambda}{4}$

$\Delta\phi = \frac{2\pi}{\lambda}.\frac{\lambda}{4} = \frac{\pi}{2}$

$I_{net} = I + I + 2\sqrt{I.I}\cos\frac{\pi}{2}$

$I_{net} = 2I$

4) $\Delta x = \frac{\lambda}{6}$

$\Delta\phi = \frac{2\pi}{\lambda}.\frac{\lambda}{6}$

$= \frac{\pi}{3}$

$I_{net} = I + I + 2\sqrt{I.I}\cos\frac{\pi}{3}$

$I_{net} = 3I$

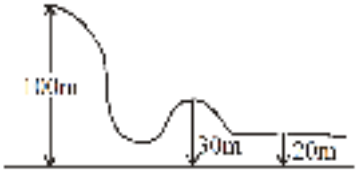
25) Heat lost = Heat gain

$100 \times 0.1(T - 75) = 170 \times 1 \times (75 - 30) + 100 \times 0.1 \times (75 - 30)$

$10(T - 75) = (170 + 10)(75 - 30)$

$10(T - 75) = 180 \times 45$

T = 885°C

26) 

Using, energy - conservation theorem

Loss in PE = gain in kE

$mg(\Delta h) = \frac{1}{2}mv^2 + \frac{1}{2}I\omega^2$

$\Delta h = 80m, \quad \omega = \frac{v}{r}; \quad I = \frac{2}{5}mr^2$

$g = 10\ m/s^2$

$v = 40\sqrt{\frac{5}{7}}\ m/s$

27) J = Iw = constant

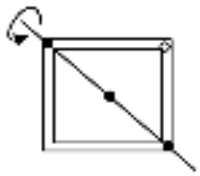
$\frac{2}{5}mR^2w = C$

$\Rightarrow w \propto R^{-2}$ (m is also constant)

but $R \propto V^{1/3}$

$w \propto V^{-2/3}$

or $\frac{\Delta w}{w} \times 100 = \frac{-2}{3}\left(\frac{\Delta V}{V} \times 100\right) = \frac{-2}{3}(3\%) = -2\%$

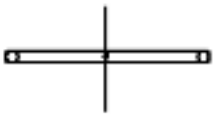
28) 

$I = 4\left(\frac{1}{3}m(L\sin 45^\circ)^2\right)$

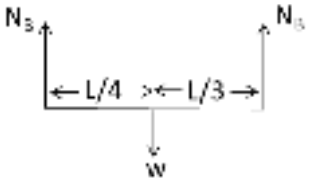
$I = \frac{2}{3}ML^2 \Rightarrow ML^2 = \frac{3I}{2}$

$= \frac{3 \times 24}{2} = 36\,kgm^2$

So,



$I' = \frac{1}{12}ML^2 = \frac{1}{12}(36) = 3\,kgm^2$

29) 

Torque about center point is zero

$$N_A \frac{L}{4} = N_B \frac{L}{3}$$

$$\frac{N_A}{N_B} = \frac{4}{3}$$

30) $v \propto \sigma^a \rho^b \lambda^c \Rightarrow [LT^{-1}] = [MT^{-2}]^a [ML^{-3}]^b [L]^c$

$\Rightarrow v \propto \sqrt{\frac{\sigma}{\rho\lambda}} \Rightarrow v^2 \propto \frac{\sigma}{\rho\lambda}$

31) Average value or mean value

$$= \frac{1.93 + 1.99 + 2.06 + 2.08 + 1.95}{5}$$

= 2.002 s = 2.00 s

Average errors

$$= \frac{|-0.7| + |-0.1| + 0.06 + 0.08 + |-0.5|}{5}$$

= 0.054 s = 0.05 s

Reading = (2.00 ± 0.05) s

32)

From COME :-

$$-\frac{GMm}{R+R} = -\frac{GMm}{R} + \frac{1}{2}mV^2$$

$$\frac{1}{2}mV^2 = \frac{GMm}{2R}$$

$$V^2 = \frac{gR^2}{R} = gR$$

$$\therefore V = \sqrt{gR}$$

33) We have,

$mg' = mg - m\omega^2 R \cos^2\theta$

for weightlessness, $mg' = 0$

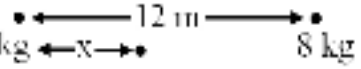
$0 = mg - m\omega^2 R \cos^2\theta$

$mg = m\omega^2 R \cos^2\theta$

$$\omega = \frac{1}{\cos\theta}\sqrt{\frac{g}{R}}$$

$$\omega = \frac{5}{4}\sqrt{\frac{g}{R}} \qquad \therefore \theta = 37°, \ \cos 37° = \frac{4}{5}$$

34)



$$x = \frac{8 \times 12}{12} = 8$$

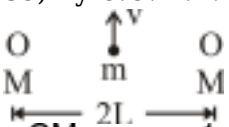
$$I_1 = \frac{G(4)}{64}$$

$I_2 = \frac{G(8)}{16}$

$I_{net} = I_2 - I_1$

$= \frac{G(8)}{16} - \frac{G(4)}{64} = \frac{7}{16}G$

35) By C.O.M.E.



$-\frac{GMm}{L} \times 2 + \frac{1}{2}mv^2 = 0$

$\frac{1}{2}mv^2 = \frac{2GMm}{L} \Rightarrow v = 2\sqrt{\frac{GM}{L}}$

36) $Y = 7 \times 10^{10}\ N/m^2$

Strain $= \frac{0.04}{100}$

$Energy = \frac{1}{2}\left(\frac{YA}{l}\right)\Delta x^2$

$Energy = \frac{1}{2}YA\left(\frac{\Delta x}{l}\right)^2 \times l$

$\Rightarrow \frac{E}{V} = \frac{1}{2} \times Y \times strain^2$

$= \frac{1}{2} \times 7 \times 10^{10} \times \frac{0.04 \times 0.04}{10^4} = 56 \times 10^2$

37)

Velocity of air above car will be more compared to inside the car. According to Bernouli theorem pressure inside the car will be more than above the top of car.

38) $R = (n)^{\frac{1}{3}} \cdot r = 2r$

$P_{ex} \propto \frac{1}{r} \Rightarrow \frac{(P_{ex})_{big}}{(P_{ex})_{Small}} = \frac{r}{R} = \frac{1}{2}$

$\Rightarrow (P_{ex})_{big} = 3$ unit

39) Terminal velocity $v_T = \frac{2r^2}{9\eta}(\rho - \sigma)g$

$\frac{v_1}{v_2} = \left[\frac{\rho_1 - \sigma}{\rho_2 - \sigma}\right] \Rightarrow \frac{0.2}{v_2} = \frac{13.5 - 1.5}{10.5 - 1.5} \Rightarrow v_2 = 0.1$ m/s

40) $\rho = \frac{m}{V} = \frac{\pi}{2 \times \frac{4}{3}\pi(0.5)^3}$

$= \frac{3}{8} \times \frac{1}{0.125} = 3gm/cc$

Now $\eta = \frac{2}{9}\frac{r^2(\rho-\sigma)}{V_T}g \quad \left[V_T = \frac{S}{t}\right]$

$= \frac{2}{9}\frac{(0.25)(3-1.2)(980)(5)}{50}$

= 9.8 poise

41) v = 2f(□₂ – □₁) = (2)(250) (97 –31)(10^{-2})

= 330 ms^{-1}

e = $\frac{\ell_2 - 3\ell_1}{2}$ = $\frac{97-93}{2}$ = 2 cm

e = 0.6 r ⇒ r = $\frac{2\,cm}{0.6}$ = 3.33 cm

42) Weight of cylinder = Up thrust due to both liquids.

V × D × g $= \left(\frac{A}{5} \times \frac{3}{4}L\right) \times d \times g + \left(\frac{A}{5} \times \frac{L}{4}\right) \times 2d \times g$,

$\Rightarrow \frac{A}{5} \times L \times D \times g = \frac{A}{5} \times \frac{3}{4}L \times d \times g + \frac{A}{5} \times \frac{L}{4} \times 2d \times g$,

$\Rightarrow D = \frac{3}{4}d + \frac{d}{2}, \Rightarrow D = \frac{5}{4}d$

43) First, calculate the least count of the vernier callipers using the relation between its scales. This value then serves as the pitch for the screw gauge, from which its own least count can be determined.

5 V.S.D = 4 MSD

1VSD = $\frac{4}{5}$ MSD

L.C. =1MSD – 1VSD = $\frac{1}{5}MSD = 1mm$

For screw guage LC = $\frac{1mm}{100}$ = 0.01 mm

44) $\frac{g\sin\theta}{g\sin\theta - \mu g\cos\theta} = \frac{2}{1}\frac{\frac{3}{5}}{\frac{3}{5} - \mu\left(\frac{4}{5}\right)} = \frac{2}{1}$

= 3 = 6 – 8 μ

8μ = 3 ⇒ μ = 0.375

45)

F = ma, where,

a = slope of v-t graph.

CHEMISTRY

46)

Element configuration will be ... $4s^2$, $4p^6$, $4d^8$ $5s^2$
Pd(46) $4s^2$, $4p^6$, $4d^{10}$ $5s^0$

47)

4th period of $4s^2$ $3d^{10}$ $4p^6$ (if 2 'e' in a orbital)
$4s^4$ $3d^{20}$ $4p^{12}$ (if $4e^-$ in a orbital)

48) ${}_3Li^+$ & ${}_5B^{3+}$ are isoelectronic hence $Z_{eff} = Z/e$
So $Z_{eff} \propto Z$

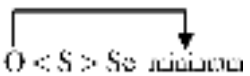
49)

Refer IUPAC naming of element from Periodic table-NCERT

50)

[Ne^+ < Ne] IP, successive IP always increase

51)

In group 16 
Hence [S > Se > O] correct order of EA.

52)

C_2 configuration :
$\sigma_{1s}^2\sigma_{1s}^{*2},\sigma_{2s}^2\sigma_{2s}^{*2},\pi_{2p_x}^2\pi_{2p_y}^2\sigma_{2p_z}^0$(2π bond) diamagnetic
if s-p mixing not allowed then :
$\sigma_{1s}^2\sigma_{1s}^{*2}\sigma_{2s}^2\sigma_{2s}^{*2}\sigma_{2p_x}^2\pi_{2p_y}^1\pi_{2p_z}^1$ (paramagnetic)

53)

XeF_4, XeF_5^-, XeF_3^+, ICl_4^- and SCl_2
total 5 species

54)

$BF_3 = CO_2 = CCl_4$, H_2O > HF and $SO_2 > SO_3$

55)

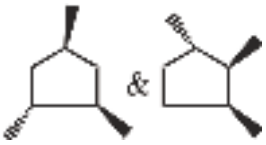
H_2O V-shape, XeF_2 linear, XeF_4 square planar, SO_2 V-shape, BrF_3 T-shape

56)

Since PH_3 lies in Drago rule hence have l.p. in pure (i.e. 100%) s- orbital

57)

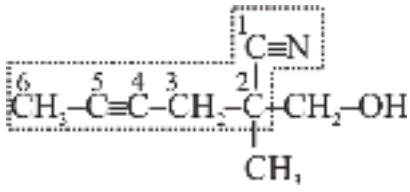
$sp^3d^3 \rightarrow {}^{\partial}x^2-y^2, {}^{\partial}z^2$ and ∂_{xy}
$CH_2=CH-C\equiv CH$
C_2-C_3 is sp^2-sp

58)  are positional isomers.

59) CH_3-CH_2-CHO : $CH_3-\underset{\underset{O}{\|}}{C}-CH_3$
Propanal Propanone ⇒ Functional isomer

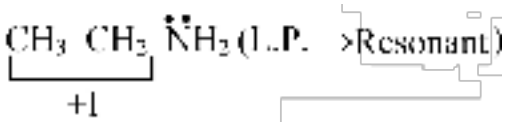
$CH_3-CH_2-O-CH_2-CH_3$: $CH_3-O-CH_2-CH_2-CH_3$
Ethoxy ethane Methoxy propane ⇒ Metamers

$CH_3-CH_2-CH_2-CH_2-CH_3$: $CH_3-\underset{\underset{CH_3}{|}}{\overset{\overset{CH_3}{|}}{C}}-CH_3$
Pentane 2,2-Dimethyl propane ⇒ Chain isomers

60)  2-hydroxy methyl-2-methyl hex-4-yne nitrile

61) i, ii, iv, vi
(Total - 4)

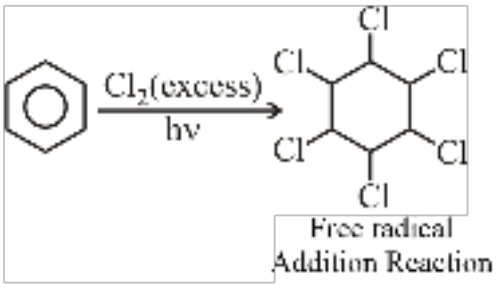
62) $CH_3-CH_2-\underset{\underset{O}{\|}}{C}-\overset{\ominus}{C}H-\underset{\underset{O}{\|}}{C}-CH_3 \rightarrow$ Most stable carbanion

63) 

64) 

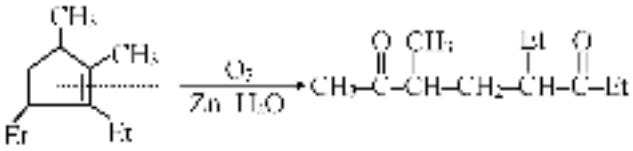
65)

Reactivity for Dehydration ∝ stability of carbocation

66) 

67)

Rate of Nitration ∝ +M , +H, +I group

68) 

69)

Only terminal alkyne give white ppt. with tollen's reagent.

70)

F.C.R. absent in Nitrobenzene (NO_2 → – M group)

71) $S^{2-} + [Fe(CN)_5 NO_2]^{2-} \rightarrow [Fe(CN)_5 NOS]^{4-}$
Violet

72) **Solution:**
Assertion (A) : Correct, because o-nitrophenol and p-nitrophenol can be separated by steam

distillation.
Reason (R) : Correct, o-nitrophenol has intramolecular H-bonding → more volatile → steam distillable. P-nitrophenol has intermolecular H-bonding → less volatile → not steam distillable.

73) $d_{H_2O(\ell)}$ = 1 g/ml
20g = 20 ml

74) $CaO(s) + H_2O \rightarrow Ca(OH)_2$
$n_{CaO} \Rightarrow \left(\frac{1000g}{56}\right)$ mol
$n_{H_2O} \Rightarrow \left(\frac{1000}{56}\right) \times 18$
$W_{H_2O} \Rightarrow 321.4$ g

75) $I_2 + I^- \rightleftharpoons I_3^-$
1 – x (0.5 – x) x
(0.5 – x) = 0.25
x = 0.25
$K = \frac{x}{(1-x)(0.5-x)} = \frac{0.25}{0.75 \times 0.25}$
$= \frac{4}{3} = 1.33$

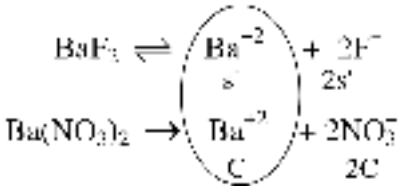
76) Equilibrium shift in forward direction by adding reactant.

77) K_{eq} → depends only on temperature for a given reaction.

78) $NH_4HS(s) \rightleftharpoons NH_3(g) + H_2S(g)$
P P
$P_T = 2P$
$\frac{3}{2} = P$
1.5 P
$K_P = P^2$
$= (1.5)^2$

79) $(H^+) = \sqrt{K_a C} = \sqrt{2 \times 10^{-12} \times 0.02}$
$(H^+) = 2 \times 10^{-7}$
Contribution of (H^+) is considered from water
$(H^+) = 2 \times 10^{-7} + (10^{-7})$←water
$(H^+) = 3 \times 10^{-7}$
pHz = 7 – log 3 = 6.52
but due to common ion effect (H^+) is slightly less than 3×10^{-7} so pH is slightly greater than 6.52.

= 6.59



80)

$[F^-] = 2s^1$

$\frac{[F^-]}{2} = s'$

$\boxed{0.5\,[F^-]}$

81) $Co(OH)_3 \rightleftharpoons Co^{+3} + 3OH^-$

s 3s

$K_{sp} = 27\ s^4 = 27 \times 10^{-44}$

$s = 10^{-11}$

$[OH^-] = 3 \times 10^{-11}$

pOH = 10.52

82) CH_3COONH_4 shows cationic as well anionic hydrolysis.

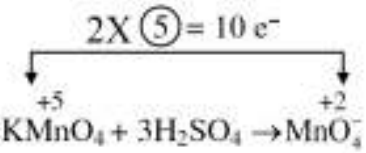
83) Fact base for isothermal T remains constant.

84) At equilibrium $\Delta T = 0$

$C_P = \frac{q}{\Delta T} = \infty$

85) Spontaneous nature depends on ΔH and ΔS not only on ΔH.

86)

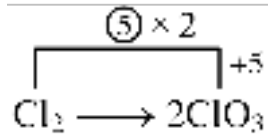


3 mole H_2SO_4 transfer $10e^-$

1 mole H_2SO_4 transfer = $\frac{10}{3}$ = v.f.

Eq. wt = $\frac{M}{\frac{10}{3}} = \left(\frac{3M}{10}\right)$

87) (1)



v.f = 10

(2) $2Cl_2 \xrightarrow{2e^-} Cl^- + ClO^-$

v.f = 1

(3) $Cl_2 \rightarrow Cl^-$

v.f = 2

(4) $Cl_2 \rightarrow Cl^- + ClO^-_3$

$v_f = \frac{5}{3}$

Eq. wt. = $\frac{71}{5/3} = \frac{71 \times 3}{5} = 42.6$

88)

Principal Based

89) g → (9) orbitals

90) Energy of an electron in single electron system depends only on (n) not (□).

BIOLOGY

91) NCERT-XI, Pg. # 111, 112

92)

NCERT-XIth, Pg. # 111, 112

93)

NCERT-XIth, Pg. # 113

94)

NCERT-XIth, Pg. # 109, 111

95) NCERT-XI, Pg. # 136

96) NCERT-XI, Pg. # 143,144,145

97) NCERT, Pg. # 138

98) NCERT, Pg. # 138,139,140

99) NCERT, Pg. # 143,144

100) NCERT, Pg. # 159,160

101)

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102)

NCERT XI Pg. # 157

103)

NCERT-XIth, Pg. # 158, 159

104)

NCERT-XIth, Pg. # 158

105) NCERT XI, Pg#230 (Fig. 14.2)

106) NCERT-XIth, Pg. # 174

107)

NCERT-XIth, Pg. # 167, 168, 169

108)

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109)

NCERT-XIth, Pg. # 174

110) NCERT-XIth, Pg. # 12, 13, 90

111) NCERT, Pg. # 18

112) NCERT, Pg. # 17,18

113) NCERT Pg. # 23, 24

114) NCERT Pg. # 14, 15, 16

115) NCERT-XIth Pg. # 78

116) NCERT-XIth Pg. # 78

117) NCERT-XII Pg # 17,18

118)

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120) NCERT, Pg. # 30, 32

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125) NCERT Pg. # 125

126) Module

127) NCERT Pg. # 121, 122

128) NCERT_Pg. # 126

129) XI-NCERT Page No. # 169, 170

130) NCERT Pg. No. # 132

131) NCERT-XI Pg#135

132) XI NCERT Page. No :- 133

133) NCERT Pg. No. 97-98

134) NCERT page No. - 96

135) Module

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137) NCERT, Pg. # 40,41,42

138) NCERT (XII) ; Page No. # 47

139) NCERT, Pg. # 47, 48, 49

140) NCERT Pg. # 48

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143) Module

144) Module

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158) NCERT, Pg. # 209-210

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172) NCERT XI Pg.#244

173) NCERT Pg # 242

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175) NCERT Pg. No. # 227

176) NCERT Pg. No. # 227

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180) NCERT Pg. # 221